# Incorporating Administrative Data in Survey Weights for the 2018-2022 Survey of Income and Program Participation

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**Abstract**

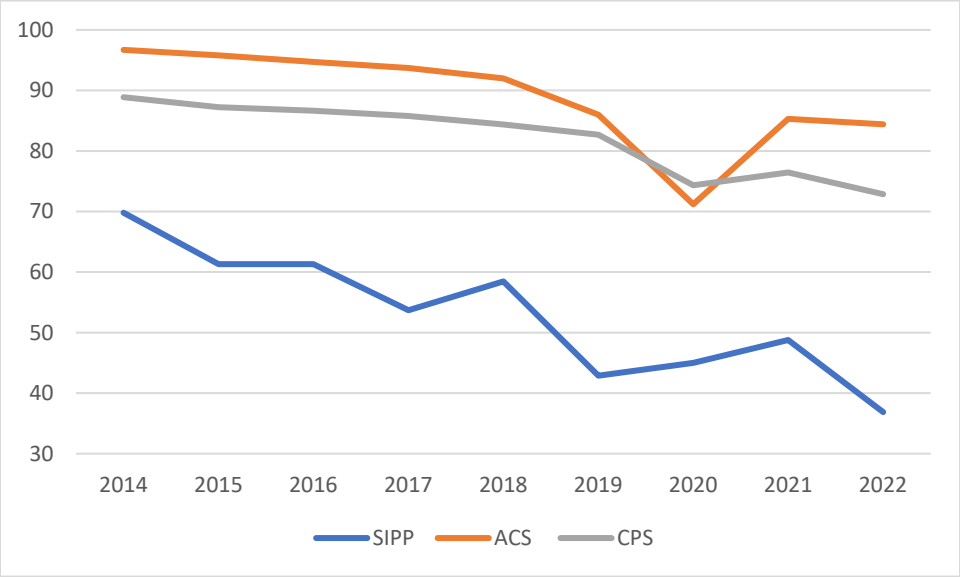
Response rates to the Survey of Income and Program Participation (SIPP) have declined over time, raising the potential for nonresponse bias in survey estimates. A potential solution is to leverage administrative data from government agencies and third-party data providers when constructing survey weights. In this paper, we modify various parts of the SIPP weighting algorithm to incorporate such data. We create these new weights for the 2018 through 2022 SIPP panels and examine how the new weights affect survey estimates. Our results show that before weighting adjustments, SIPP respondents in these panels have higher socioeconomic status than the general population. Existing weighting procedures reduce many of these differences. Comparing SIPP estimates between the production weights and the administrative data-based weights yields changes that are not uniform across the joint income and program participation distribution. Unlike other Census Bureau household surveys, there is no large increase in nonresponse bias in SIPP due to the COVID-19 Pandemic. In summary, the magnitude and sign of nonresponse bias in SIPP is complicated, and the existing weighting procedures may change the sign of nonresponse bias for households with certain incomes and program benefit statuses.

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project P-7521191: CBDRB-FY24-CES005-012). All comparative statements in this report have undergone statistical testing, and, unless otherwise noted, all comparisons are statistically significant at the 10 percent significance level. Statistics from surveys are subject to sampling and nonsampling error. For further information on the source of the data and accuracy of the estimates, including standard errors and confidence intervals, see http://www.census.gov/programs-surveys/sipp/tech-documentation/source-accuracy-statements.html.

# 1. Introduction

Declining unit response rates in surveys are a major concern for survey data providers and users. In addition to surveys from academic institutions and the private sector, a variety of surveys administered by statistical agencies are also experiencing this long-run trend (Czajka and Beyler 2016). Moreover, the COVID-19 Pandemic had a unique, acute effect on data collection when in-person interviewing was suspended for Census Bureau surveys starting on March 20, 2020 and continued to be suspended or highly restricted during much of 2020 (Rothbaum et al. 2021). Hiring difficulties and shortages of interviewers have also affected Census Bureau surveys during the early part of the 2020s.

Figure 1: Response Rates for Surveys Conducted by the U.S. Census Bureau




*Source: U.S. Census Bureau, 2014-2022 Survey of Income and Program Participation https://www.census.gov/programs-surveys/sipp/tech-documentation/nonresponse-reports.html; U.S. Census Bureau, 2014-2022 American Community Survey https://www.census.gov/acs/www/methodology/sample-size-and-data-quality/response-rates/; U.S. Census Bureau, 2014-2022 Current Population Survey https://www.bls.gov/osmr/response-rates/household-survey-response-rates.htm*

Figure 1 shows annual response rates for the Survey of Income and Program Participation (SIPP), the American Community Survey (ACS), and the Basic Monthly Current Population Survey (CPS). ACS and CPS have declining response rates over time, with a notable decrease in 2020. SIPP's response rates are lower and more volatile over time, the reasons for which are discussed later in the paper. These lower response *rates* may increase the potential for nonresponse *bias* in SIPP estimates. Note that it is not necessarily true that nonresponse bias is increasing from these rising nonresponse rates, as

nonresponse rates are only weakly associated with nonresponse bias in surveys (Groves and Peytcheva 2008).

In this paper, we propose improvements to the SIPP weighting algorithm that leverage administrative data from federal agencies, state agencies, and third-party providers to address possible nonresponse bias in SIPP. Specifically, we match income data from IRS tax forms, program benefit data from various federal and state agencies, and demographic data from the Social Security Administration and the 2010 Decennial Census to SIPP sampled units, both respondents and non-respondents. We then use this matched data in the household nonresponse adjustment part of the SIPP weighting algorithm, which adjusts the weights of respondents to account for differential nonresponse rates among subpopulations and reduces nonresponse bias in survey estimates. We also add administrative data to the raking step of the weighting process, where we calibrate to age, race, and sex data from Census Bureau's Population Estimates program and means of administrative data variables. We use the full ACS sample – as opposed to the full SIPP sample – matched to administrative data to create these additional calibration target moments to improve the precision of our estimates. Additionally, many nonresponding SIPP households have unknown vacancy status due to shortages of interviewers and suspension of in-person interviewing during 2020, whereas relatively fewer ACS nonresponding households have unknown vacancy status.

Administrative data has been used for survey weights in research on Census-conducted and Census-sponsored surveys in recent years. Such research has been performed on the ACS (Rothbaum et al., 2021), the Basic Monthly CPS (Eggleston et al. 2024), the CPS Annual Social and Economic Supplement (Rothaum and Bee, 2021), the CPS Food and Security Supplement (Eggleston et al. 2023), the Consumer Expenditure Survey (Ash and Rothbaum, 2024), and the 2014 Survey of Income and Program Participation (Eggleston and Westra, 2020). This paper largely extends this prior work to analyze additional years of SIPP. Compared to Eggleston and Westra (2020), this paper also uses additional administrative data on program participation and modifies the calibration step, not just the household noninterview adjustment.

We create these new weights for 2018 through 2022 SIPP and examine how the new weights affect survey estimates and the balance of the respondent sample as measured through the administrative data. Our results show that before weighting adjustments, SIPP respondents in these panels have higher socioeconomic status than in the ACS benchmark. Existing SIPP weighting procedures reduce many of these differences. Comparing the SIPP estimates between the production weights and

the administrative data-based weights yields changes that are not uniform across the joint income and program participation distribution. For example, in 2022 SIPP, the administrative records (adrec)-based weights yield both higher estimates of poverty and lower estimates of SNAP receipt, which is somewhat surprising given poverty status and SNAP receipt are positively correlated with each other.

We describe the magnitude of the changes as "moderate," although this is open to individual interpretation. For example, with administrative data-based weights, the estimate of the poverty rate increases by 0.26 percentage points for 2021 SIPP and increases by 0.43 percentage points for 2022 SIPP. The change in the estimate of median household income is between $500 and $1,000 for 2018, 2019 and 2021 SIPP, and the change is insignificant for 2020 and 2022 SIPP. Some years yield a larger change for some estimates. For example, the estimate of Medicaid receipt in 2019 SIPP decreases by 0.69 percent with administrative data-based weights, while the analogous decrease was insignificant for 2021 SIPP.

In summary, using administrative data shows a complicated picture on both the magnitude and direction of nonresponse bias in 2018-2022 SIPP. It appears that while the existing weighting methods reduce nonresponse bias, the adjustments may overcorrect for households with certain incomes and program benefit statuses. By overcorrect, we mean that before weighting adjustments, SIPP respondents have higher income than the target population. But then for some statistics and SIPP years, weighting adjustments lead to a lower estimate of income relative to the population target. Thus, these weighting adjustments can change the direction of nonresponse bias. While the magnitude and direction of nonresponse bias varies by year and topic, it does not appear that any one year of SIPP has uniformly higher nonresponse bias than another year for all variables even with the unique effects of COVID-19 on data collection in 2020. This is in contrast with the ACS and CPS Annual Social and Economic Supplement, which saw a notable increase in nonresponse bias due to the COVID-19 Pandemic.

# 2. Data

## 2.1 Overview of the Survey of Income and Program Participation

The Survey of Income and Program Participation is a nationally representative panel survey administered by the Census Bureau. It collects information on the short-term dynamics of employment, income, household composition, and eligibility and participation in government assistance programs among the civilian, noninstitutionalized population of the United States. It is a leading source of

information on specific topics related to economic well-being, family dynamics, education, wealth and assets, health insurance, childcare, and food security. Each SIPP panel follows individuals for several years in interview cycles called waves, providing monthly data that measures changes in household and family composition and economic circumstances over time. To obtain more precise estimates of participation in government programs, SIPP oversamples households in high-poverty areas.

We analyze the 2018 through 2022 SIPP survey years. These years refer to the year a household was interviewed, where the reference year is the prior year (e.g., in 2018 SIPP, households were interviewed in 2018 and asked about income primarily from 2017). In these SIPP survey years, households are interviewed for up to 4 years unless they attrite. 2018 SIPP contained an entirely new set of households who had not been interviewed before. 2019 SIPP brought about the return of the overlapping panels, which were present in SIPP until the 1996 Panel.[2] In 2019 SIPP, a new set of households were interviewed along with returning households from 2018 SIPP. The interview cohort is referred to as the *panel*, with the year referring to the *first* year a household was interviewed. 2019 SIPP contained households from both the 2019 Panel (interviewed for the first time in 2019) and the 2018 Panel (interviewed for a second time in 2019). During this time period, the mode of response was either in-person interviews or by phone.

These years of SIPP were marked by various interviewing challenges. As Figure 1 shows, SIPP has experienced overall declining response rates over time, with 2014 SIPP Wave 1 having a response rate of 69.8 percent but the 2018 SIPP having a response rate of 58.44 percent. From 2019 onward, additional interviewing difficulties were present as well. The 2018-2019 lapse in federal appropriations led to delays in hiring and training interviewers for Census Bureau surveys. This led to the 2019 SIPP interview cycle starting with insufficient staff. To help maintain overall data quality over time, a decision was made to deprioritize the 2019 Panel (Wave 1) cases during collection to focus effort on maintaining the 2018 Panel (Wave 2) cases. In 2020 SIPP, the COVID-19 Pandemic resulted in the suspension of in-person interviewing on March 19, 2020, so telephone became the only mode of response. 2021 and 2022 SIPP were also impacted by interviewer shortages affecting the entire Census Bureau during the early 2020s.

[2] The 1993 Panel was the last panel under the prior overlapping panel regime.

Table 1: Response Rates for 2018 to 2022 SIPP by Panel

| | 2018 SIPP | 2019 SIPP | 2020 SIPP | 2021 SIPP | 2022 SIPP |
|---|---|---|---|---|---|
| Wave 1 | 58.44 | 25.35 | 35.99 | 42.49 | 30.74 |
| Wave 2 | | 60.76 | | 63.54 | 54.68 |
| Wave 3 | | | 52.28 | | 48.56 |
| Wave 4 | | | | 47.74 | |

Source: https://www.census.gov/programs-surveys/sipp/tech-documentation/nonresponse-reports.htmls Note that the 2019 Panel was dropped from further data collection after Wave 1 due to low response rates.

To help summarize these effects, Table 1 shows response rates for SIPP for 2018 to 2022. The table presents the weighted cross-sectional response rate for each wave. The response rate is highly variable by year and panel. Due to reasons discussed above, the 2019 Panel Wave 1 had a response rate of only 25.35 percent. With these response rates low for a variety of reasons, there is a potential for nonresponse bias in SIPP that merits investigation.

## 2.2. Administrative Data

In this section, we describe the administrative data we match to SIPP as well as the ACS (reasons for using the ACS are discussed in Section 2.4). Our primary administrative data consist of annual tax data from the Internal Revenue Service (IRS). First, we use IRS Form 1040 individual income tax returns. The 1040 returns include income from the prior year and the identities of the first four children or other family members claimed as dependents. Next, we use IRS information returns, which include W-2s from employers, 1099 forms such as the Form 1099-INT, and Form 1098. These forms give us additional information on a variety of sources of income, as well as broader economic and financial activity. Receipt of a 1098 from a mortgage lender indicates whether the household has a mortgage, a proxy for home ownership. Filing status from a 1040 return also gives us a proxy for marital status.

To match survey and administrative records at the address level, we utilize the linking identifier in the Master Address File (MAF), the Census Bureau's frame file of all known living quarters and certain nonresidential addresses in the United States. Since 2014, the MAF has been used as the sampling frame for the SIPP, so all SIPP observations have this linking identifier, called a MAFID. The IRS data are linked to the MAF using a probabilistic linking algorithm. Looking at 1040 data, Bee et al. (2015) find that about 90 percent of Form 1040 records match to a MAFID, with unmatched tax records tending to be in the extreme upper and lower ends of the income distribution. There are two possible reasons for why a

household does not match to IRS data. It could be that this household did not file their taxes or receive any other tax form (among those provided to the Census Bureau) from an employer or other party. Or it could also be that they did have one of these tax forms but a problem with the address listed on their tax form resulted in no MAFID being assigned. As mentioned in Rothbaum et al. (2021), while this non-linkage makes the reweighting less efficient, it should remain consistent if the relationship of non-linkage to the data is independent of survey response, which is an open empirical question.

### 2.2.1 Construction of Household Roster

After linking several administrative datasets to the SIPP by address (MAFID), our second step is to use the administrative data to construct a household roster for each address that we then use as a basis for matching additional person-level datasets. Our primary sources for constructing a household roster are the IRS 1040 returns and the IRS information returns (e.g., Forms 1098, 1099, and W-2). Importantly, the dependent data in the 1040s gives us information about children in the household. Using the information returns in addition to the 1040s is useful for picking up individuals who do not have any filing requirements. For example, retired individuals who only receive Old-Age, Survivors, and Disability Insurance (OASDI) benefits typically do not have any taxable income. These individuals then might not file an IRS 1040 tax return, but they would receive an SSA-1099, allowing us to add them to household rosters.

However, some people in the United States do not appear in any IRS data available to the Census Bureau. To capture some of these individuals, we utilize two additional sources when constructing the household rosters. First, we use location information from Census Bureau's Master Address File Auxiliary Reference File (MAFARF). The MAFARF contains residency information from IRS data and other administrative data sources, such as the United States Department of Housing and Urban Development (HUD). Second, we use data on Supplemental Security Income (SSI) recipients from SSA's Supplemental Security Record (SSR). Since SSI is not taxable, individuals whose sole source of income is SSI would not appear in IRS data. Finally, we use parent-child linkages from the Census Bureau Household Composition Key to add children to the household roster who do not appear in any of our other sources. This file is created using data Census receives from SSA on parent names (but not SSNs), confirming that the parent and children live together using other administrative data or a decennial census (see Genadek et al., 2021 for more details).

The household roster we create from the administrative datasets will likely not match exactly the household roster for some SIPP respondent households (see Brown et al., 2023 for a discussion of

this issue with the 2020 Census). However, our reweighting exercise does not require us to replicate exactly the household roster that appears (or, for nonrespondents, would appear) in SIPP. Instead, we merely need to construct a *proxy variable* of who resides in each household. The more closely correlated these proxies variable are to dimensions of the data that are correlated with response, the more efficient the weighting becomes. For example, consider a child of divorced parents. A custody agreement may result in a difference between which parent claims the child on the 1040 in a year and where the child is deemed to reside in accordance with SIPP residency rules. However, if these discrepancies are similar between respondent and nonrespondent households, the reweighting should remain consistent. To the extent that these discrepancies are relatively rare, they result only in small reductions in how well our administrative data variables are correlated with SIPP survey data, and thus the efficiency of the reweighting process.

### 2.2.1 Additional Data Linkage

We also link several additional datasets to learn more about these households. We use demographic data from the SSA's Numident file and the 2010 Decennial Census. Both the Numident and 2010 Census data contain information on an individual's age and race. If race data are available from both sources for a person, we use the decennial census race. The 2010 Census also provides information on Hispanic origin, while the Numident data contain information on citizenship and foreign-born status.

For program benefit information, we link Medicaid data from the Centers for Medicare & Medicaid Services to obtain data on participation in Medicaid and the Children's Health Insurance Program. In addition to the SSR described earlier, we match OASDI program benefit data for SSA's Master Beneficiary Record (MBR) and the Payment History Update System (PHUS). We also link benefits records from state data for the Supplemental Nutrition Assistance Program (SNAP); the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC); and Temporary Assistance for Needy Families (TANF). Two notes need to be made about these data. First, the U.S. Census Bureau only has data acquisition agreements for certain states, so there is incomplete national coverage of the SNAP, WIC and TANF data. Additionally, to simplify our data cleaning procedure and to make our code more robust to layout changes in the SNAP, TANF, and WIC data in the future, we only examine whether a person is present in these data and do not make use of additional information to confirm they are receiving benefits in a given year. With this simplification, we may be flagging a small number of individuals as receiving SNAP, WIC or TANF who may either no longer be receiving these benefits or applied for but were denied these benefits.

These datasets are linked to the IRS-based household rosters at the individual level using the Census Bureau's Person Identification Validation System (PVS), as described by Wagner and Layne (2014). This procedure matches both survey and administrative data to a master reference file. Individuals who are matched are given an identifier called a Protected Identification Key (PIK), which acts as an anonymized Social Security number that can be used to link administrative datasets and surveys. For the 2010 Census, about 90 percent of individuals were matched to a PIK (Wagner and Layne, 2014).[3] Bond et al. (2014) argue that inability to match to a PIK is nonrandom. They use 2009 ACS data to document that young children, minorities, immigrants, recent movers, low-income individuals, and non-employed individuals are less likely to be matched to a PIK. PVS rate are much higher for the other administrative datasets, given they contain Social Security Numbers.

Once the Numident, 2010 Census, Medicaid, SSA SSI and OASDI, SNAP, TANF, and WIC data are matched at the individual level, the data are then aggregated to the household level to create household-level measures for comparing respondent and nonrespondent households. For example, we create an indicator for the presence of a household member over age 60 by taking the list of people on the tax forms matched to an address and matching these people to the Numident file to get the year of birth for all household members.

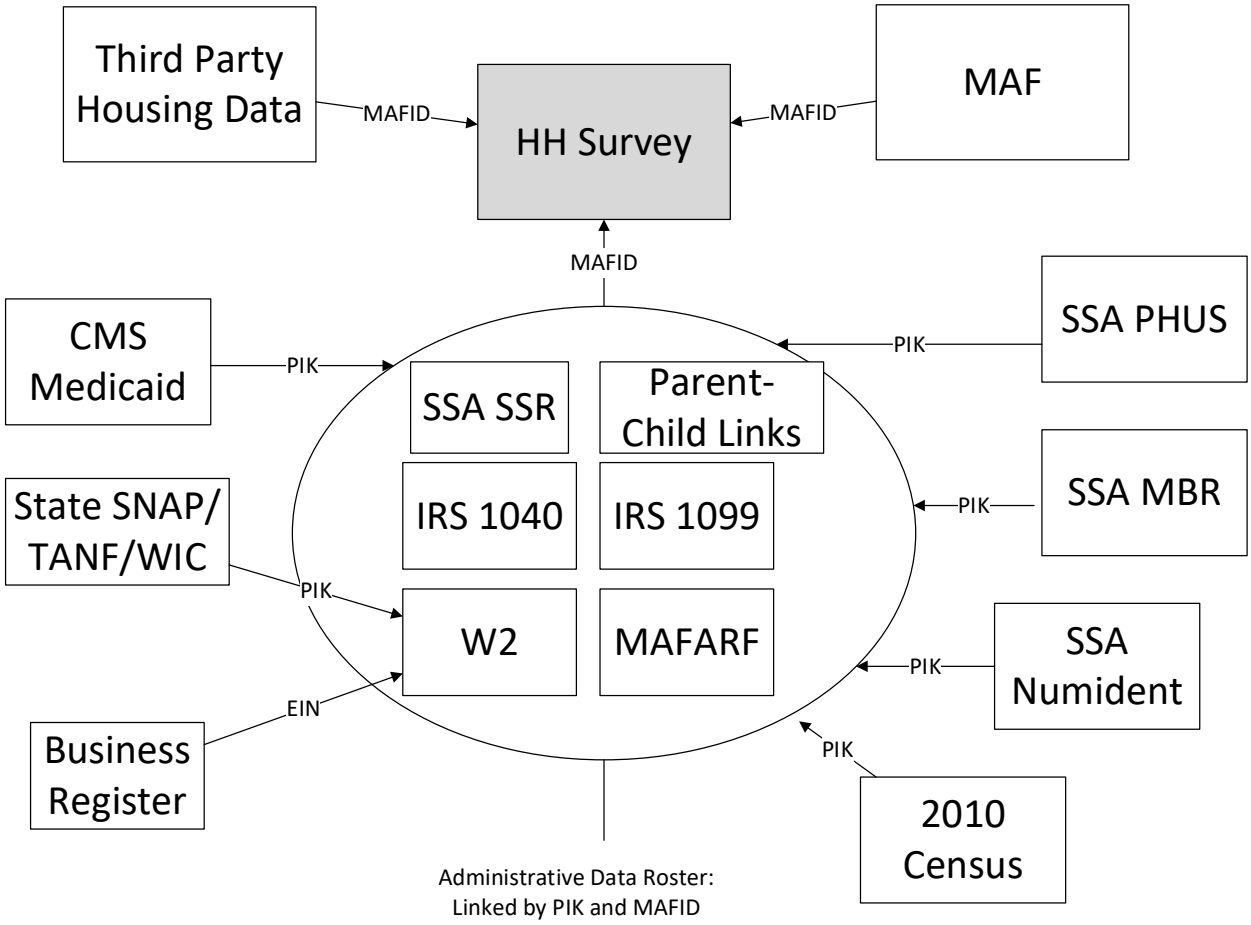
To obtain information about employer characteristics, we use the Employer Identification Numbers (EIN) on the W-2s matched to the household. First, we leverage the W-2s of a household member's coworkers to obtain a proxy of how many people work at their employer in case there are unemployment shocks in a month that are correlated with firm size. Second, we use the Census Bureau's Business Register to obtain information on the industry in which each household member works. Given that the employment of people in the hospitality and food service industry was particularly affected by COVID-19 restrictions, this industry information may be especially valuable for economic outcomes in 2020. Note that this linkage by EIN is imperfect for many businesses, as EIN is neither a firm nor an establishment identifier for multi-establishment firms. In other words, the number of people who share the same EIN may not always equal the number of people that work at a person's place of business. However, since our goal is to construct proxies for employer characteristics rather than to

[3] The 2010 Census is also linked to the MAF. Because many people have changed their residential location since 2010, however, we link decennial data at the person level instead, to make sure we are capturing the characteristics of the current household.

construct precise establishment-level statistics, and we do not expect the validity of these proxies to vary by survey response, we believe the EIN linkage is adequate as an input for our weighting algorithm.

Finally, we link commercial housing data from Black Knight, a third-party vendor, and structure-type information from the MAF. From the Black Knight data, we use information on the home's assessed value as well as whether it is owner-occupied. The home's value and whether the residence is a single-family home or in a multi-unit structure provide information on the socioeconomic status of households, particularly for households that we are unable to match to any other administrative data. Figure 2 summarizes graphically all administrative data linkages.

**Figure 2: Data-Matching Diagram**




## 2.3 Mover Addresses

In many of our analyses, we want to match administrative data to the address a household is at in the current interview year. SIPP tracks movers to their new addresses. Unfortunately, MAFID are not assigned to the new mover addresses in current SIPP data processing. For households that move, we take instead the MAFID from the MAFARF. For each person in the mover household who is assigned a PIK, we match this PIK to the MAFARF to get the MAFID for that person. We then take the modal MAFID as the new mover MAFID. If no person has a PIK or no person in the mover household is matched to a MAFID, then we still use the original sample MAFID for merging administrative data to that calendar year. In the future, if mover addresses were assigned a MAFID in SIPP data processing, then this step would no longer be needed.

## 2.4 American Community Survey

We also match administrative data to the American Community Survey. The ACS is the largest household survey in the United States, sampling approximately 3.5 million addresses each year, divided into twelve monthly panels for continuous interviewing. It is a multimodal survey where most households are given two months to self-respond by web or mail, followed by in-person follow-up in the third month for a subsample of the panel. The survey contains questions on a variety of social, housing, and economic topics, and is the key source of such statistics at the county or subcounty level in the United States. When we use the ACS to create a representative sample of occupied housing units (described in more detail in Section 3), we exclude vacant, deletes, and all addresses not sampled for CAPI. The weights we use for creating ACS-based administrative data statistics are the base weight times the CAPI subsampling factor for addresses sampled for CAPI. For more details on some of these nuances of the ACS, see Appendix A in Rothbaum et al. (2021).

We use the ACS for several reasons. First, the ACS has a much larger sample size than SIPP, giving a more precise estimate of means of administrative data variables in our modified SIPP weighting procedures. Additionally, using the ACS allows us to address issues SIPP had with unknown vacancy status of addresses. In SIPP, vacant and nonexistent sample addresses are excluded from most data processing, since all household-level statistics in SIPP are for occupied housing units. For matching administrative data to SIPP, we ideally want to exclude these addresses as well. However, given the interviewer shortages affecting SIPP in 2019 and beyond, for addresses where no response was received, occupancy status is unknown for many of these addresses. In data processing, these unknown addresses are treated as nonrespondents in occupied housing units, which may affect some of our analyses.

Table 2: ACS Sample Months Used for Constructing Administrative Data Means.

| SIPP Year | ACS Sample Months |
|---:|---:|
| 2018 | December 2017-February 2018 |
| 2019 | January 2018 - December 2018 |
| 2020 | January 2019 - December 2019 |
| 2021 | November 2019-January 2020 |
| 2022 | January 2018 - December 2018 |

This issue also affected ACS in 2020 when in-person interviewing was suspended, but we can restrict our analyses to only the sample ACS months when in-person interviews occurred and where an

interviewer could verify the vacancy status of all addresses. Table 2 above shows, for each SIPP year, the ACS sample months we use as a comparison. For 2018 SIPP, we currently only use a few months of sample given data availability restrictions in our project space. Note that we only match administrative data to ACS housing units, while our SIPP addresses include some non-institutional group quarters such as college dormitories. For 2018 SIPP, 1.26 percent of in-scope living quarters were group quarters observations, so this universe difference should have a marginal impact on any subsequent analyses. For group quarters observations, given many people may share the same MAFID, we rescale each match of our group quarters data to resemble a single-person household. For example, if a SIPP group quarters observation was at a MAFID where 10 individuals filed a tax return, we would sum up the income from all the tax returns and divide it by 10. This allows the adrec data for group quarters observations to resemble a person residing by themselves in an apartment, which is similar to someone having their own room in a group quarters structure.

Table 3: Match Rates to IRS 1040

| SIPP Interview Year | SIPP Wave 1 Benchmark Estimate | S.E. SIPP | ACS Benchmark Estimate | S.E. ACS |
|---|---|---|---|---|
| 2018 | 80.27 | 0.1912 | 80.21 | 0.0829 |
| 2019 | 72.42 | 0.2861 | 79.39 | 0.0433 |
| 2020 | 73.3 | 0.2892 | 81.54 | 0.0365 |
| 2021 | 73.15 | 0.3449 | 80.76 | 0.0668 |
| 2022 | 71.81 | 0.2524 | 80.36 | 0.0387 |

Source: ACS, SIPP, and IRS 1040 data. Table presents match rates to IRS 1040 tax return. The SIPP benchmark includes respondents and nonrespondents in Wave 1, which may include vacant addresses in 2019 SIPP and beyond. The ACS benchmark includes a sample of ACS respondents and nonrespondents in occupied housing units. Weights used are SIPP base weights and ACS base weights times CAPI subsampling factor.

To help highlight the benefits of using ACS for administrative data means, Table 3 shows the percent of respondents and nonrespondents that are matched to an IRS 1040 record. Comparing SIPP respondents and nonrespondents (which includes some nonrespondents with unknown vacancy status) compared to the ACS benchmark (where restrictions are made to ensure vacancy status is known for all addresses), in 2018, there is no statistically significant difference between the SIPP and ACS-based estimates, with both sources showing a match rate of around 80.2 percent. However, the match rates diverge in 2019 and beyond, with the SIPP estimate being 7-8.5 percentage points lower than the ACS-based estimate depending on the year. This is likely due to many of the SIPP nonresponding addresses

with unknown vacancy status being vacant in truth, so they do not have any households living there that would have used the addresses as their filing address. Therefore, using the ACS estimate will likely help deal with the unknown vacancy issue affecting SIPP data.

## 2.5. Reference Period

The reference period of these data primarily corresponds to the reference period of the SIPP year in question. For example, for 2018 SIPP, the administrative data primarily has a reference year of 2017. There are a few exceptions and modifications to note. First, if a person does not have a tax return from the year in question but does have one from the prior year, then we use the prior year's tax return instead. This helps capture people that may be below the filing requirement one year. We also examine a person's W-2s from the prior year to see if they worked for the same employer or not to provide additional information about potential job volatility. The version of the Business Register we had access to included data up to the 2019 vintage at the time the statistics in this report were calculated, so the analyses on the 2021 and 2022 SIPP Panels use industry data from 2019 instead, although the W-2s used for these people still come from tax years (reference year) 2020 and 2021, respectively. Finally, the SSA SSI and OASDI data for 2018 and 2019 SIPP actually come from 2019 reference year administrative data, given gaps in the availability of such data in the 2010s. Note that the data used here are universe extracts of every individual in the U.S. These data are covered under separate agreements than the SSA extracts of ACS, CPS, and SIPP respondents the U.S. Census Bureau also receives, which have more coverage over time. Note that the latter extracts exclude survey nonrespondents, so that is why the universe data is needed instead.

## 2.6. Data Availability and Production Schedule

Table 4: Administrative Data Availability

| Dataset | Availability During SIPP Interview Year |
|---|---|
| Business Register | One year lag |
| IRS 1040 and 1099 | Monthly deliveries starting in February |
| MAF-ARF | One year lag |
| Medicaid Data | One to two year lag |
| SSA MBR, SSR, and PHUS | Generally available by September |
| SSA Numident | Quarterly deliveries |
| State SNAP and TANF | By spring or summer for many states |
| State WIC | One year lag |
| Third-Party Housing Data | Generally available by September |
| Parent-Child Links | February |

Finally, we discuss when the administrative data described above becomes available. As of the time this paper is written, SIPP interviews respondents primarily between February and June, followed by data processing occurring through the remainder of the year and sometimes into the following year. SIPP data processing occurs with multiple runs of the full data editing and weighting procedures, so rerunning the weighting program multiple times throughout the year before the final data release to the public is already typical. For incorporating administrative data into weighting, it would be ideal that some of the data would be available for use by July, followed by December being the final time to incorporate any remaining administrative datasets.[4] For recent SIPP panels, the data have been released in the summer after the interview year. 2023 SIPP was released in June 2022, for example. In the following paragraphs, we describe when the administrative data would be available corresponding to the SIPP reference year in question.

For all these datasets, it is important to clarify that these are universe extracts, reflecting data on the entire U.S. Population. Because of this, if there is a delay in receiving any of these administrative data, the prior vintage can always be used. For example, if there was a delay in receiving the SSA SSR data on SSI receipt, the Census Bureau could use the vintage from the prior year. While this would miss information on people transitioning in and out of SSI, the prior year data should be very informative given people typically are on SSI for a long period of time (Rupp and Scott 1995). This is different than much of the administrative data used for SIPP's model-based imputation procedures. For much of the SSA data used in these procedures, the Census Bureau uses extracts that are specific to SIPP respondents. This requires the Census Bureau to send the list of new SIPP respondents to SSA before receiving these extracts. If there are delays in these data, then data is completely missing for a new panel. Thus, using universe administrative data offers better protection against delays in administrative data receipt.

For weighting, the most important datasets are the IRS data and the SSA Numident. These will be available by July. The Census Bureau gets monthly IRS deliveries starting in February. The IRS datasets are updated throughout the tax filing season. By July, well after the April filing deadline for IRS 1040s for most people, the majority of the population's tax returns would be available. The main exception is that a sizable proportion of high-income individuals request extensions and don't submit their taxes until the

[4] This timeline assumes there is some window of time needed between the cutoff of adding updated administrative data and when the data editing and weighting programs are run for one last time before release. Such a window is likely needed for quality control purposes.

fall. Our procedure described in Section 2.5 would allow us to use the IRS 1040 data of these people from the prior year. The SSA Numident is delivered quarterly, and any delays would have minimal impact since a prior vintage could always be used.

The availability of the program benefit data is more variable. The SSA MBR and SSR are generally available by September. Thus, initial data processing would likely need to use data from prior years, but updated data would likely be available before the final run. The Medicaid data will likely be available with a one or two-year lag (relative to the reference year), so the administrative data we will have on Medicaid receipt will be from the year prior to the SIPP reference year. The SNAP data should be available for many states by the summer. The WIC data has more delays. Currently, the Census Bureau delays sending the WIC data through PVS to give time for newborns to appear in the master reference file. This is to greatly improve the linkage rate for newborns and infants in the WIC data.

For other datasets, the third-party housing data is typically sent through the PVS and MAF matching by September to match the production needs of the American Housing Survey. Using lagged versions in the initial runs should be fine given someone's relative home value should be stable from year to year. The version of the Business Register used in this research has a lag of one to two years. This would only result in missing the industry information of some newly formed companies, so the overall effect on the weighting procedures would be minimal.

# 3. Weighting Methods

This section describes how we modify SIPP weighting procedures to incorporate administrative data. We first provide an overview of the existing algorithm, and then describe how we modify key parts.

## 3.1 Existing SIPP Weighting Procedures

The goal of SIPP's weighting procedures is to produce unbiased estimates that are representative of the U.S. civilian non-institutionalized population. The basic components of the SIPP weights are:

1. A base weight that reflects the probability of selection for a sample unit.
2. An adjustment for subsampling within clusters at group quarters.
3. An adjustment for movers (in Waves 2 and beyond).
4. A nonresponse adjustment to compensate for sample nonresponse.
5. An adjustment for combining sample cases from multiple panels (if overlapping panels cover the reference period).

6. A poststratification (second-stage calibration) adjustment to correct for deviations from known population totals.

In this research, we focus on modifying items 4 and 6. In the paper, we refer to #4 as the household noninterview adjustment, and #6 and either the second-stage calibration step or the raking step. The nonresponse adjustment consists of two separate modules for initial Wave 1 nonresponse and attrition in waves 2 and beyond. For the Wave 1 nonresponse adjustment, the current SIPP weighting procedure uses the following six household characteristics for defining the nonresponse adjustment cells in Wave 1: Census Region, Metropolitan Statistical Area (MSA)/Place Status (MSA-central city, MSA-non-central city, other place), poverty stratum (high poverty, low poverty), race of reference person (Black, not Black), household tenure (owner, renter), household size (1, 2, 3, 4+ people). The first three variables are known from the sampling frame, while the latter three are obtained from the survey data for respondents and the interviewer's guess of these characteristics for nonrespondent households. The nonresponse adjustment entails calculating the weighted response rate within each cell and multiplying a responding household's weight by the inverse of this probability.

For the Wave 2 and beyond nonresponse adjustment for attrition, survey data from Wave 1 is used to create an adjustment with more detailed information. The list of variables includes tenure (owner/renter occupied), household type (female, no spouse present; age 65+; other), race and Hispanic origin of the household reference person, educational attainment of the household reference person, household size, poverty status, type of income, type of financial assets, census division, and number of imputed items. Note that these nonresponse adjustments are performed separately by panel. In 2022 SIPP, for example, the weighting procedures performed a Wave 1 nonresponse adjustment on 2022 Panel members, and then separate attrition adjustments for the respondents left in the 2020 and 2021 Panels.

Finally, the person weights go through a second stage calibration. This adjustment brings the sample estimates into agreement with independent monthly estimates of the Census Bureau's Population Estimates Program. The characteristics used for calibration include age, race, sex, Hispanic origin, and state. A spouse equalization is also included to ensure spouses have the same weight. A raking procedure is used to ensure that the weights agree with all the control totals included for calibration. This raking procedure alternates between the spouse equalization and calibrating to the Census Bureau's population estimates until convergence is achieved.

The existing SIPP weighting process includes many variables that are helpful in reducing nonresponse bias in the estimates. In particular, the inclusion of an area's poverty rate in the adjustment may be particularly helpful given how important income and program benefit statistics are to the survey. Nevertheless, adding additional information that is at the household level may yield better corrections for nonresponse bias. For example, having a measure of a household's income reported in IRS data may have a higher correlation with their SIPP income than the indicator of whether they are in the high-poverty stratum. Given weighting adjustments are improved by adding variable that have a higher correlation with survey outcomes of interest (Little and Vartivarian 2005), these administrative data inputs may improve SIPP data quality. How these administrative data inputs are incorporated is discussed in the following subsection.

## 3.2 Modification to Household Noninterview Adjustment

For the household noninterview adjustment, we replace the existing noninterview adjustment with one based on administrative data. The goal is to create new cells based on administrative data that are better able to address nonresponse bias in SIPP. Section 2.2 describes the numerous administrative data we have available for nonresponse adjustments. The large number of variables we have available makes simple cross-tabulation to create the new cells infeasible. Thus, the administrative data needs to be collapsed in some way to create a manageable number of adjustment cells.

When creating these cells, it is important that the variables used are correlated with both response propensities and the survey outcomes of interest. Little and Vartivarian (2005) show that if this is the case, then weighting with such a variable will reduce both bias and standard errors. However, if a weighting variable is only correlated with response but not the survey outcomes, then adding this variable will have no effect on bias while increasing standard errors. Because of the importance of using variables that are correlated with employment outcomes, we use a classification and regression tree (CART) model to help collapse the administrative data. CART is a nonparametric model used to predict the value of a dependent variable from several explanatory variables. Because it is nonparametric, it easily incorporates interactions, which may be important for predicting income and program participation (e.g., income interacted with presence of children). One particular benefit of CART is that it utilizes a pruning procedure to prevent overfitting.

To operationalize CART for SIPP weighting, we create a household-level recode of income and program participation as the dependent variable. Using the SIPP survey data, this household-level code is defined as follows:

- 1 if anyone in household received SSI.
- 2 if anyone in household received WIC and no one received SSI.
- 3 if anyone in household received SNAP and no one received SSI or WIC.
- 4 if anyone in the household received Medicaid and no one received SSI, WIC, or SNAP.
- 5 if household income was less than the 25$^{th}$ percentile, and no one received SSI, WIC, SNAP, or Medicaid.
- 6 if household income was between the 25$^{th}$ and 50$^{th}$ percentiles and no one received SSI, WIC, SNAP, or Medicaid.
- 7 if household income was less than the 50$^{th}$ and 75$^{th}$ percentiles and no one received SSI, WIC, SNAP, or Medicaid.
- 8 if household income was greater than the 75$^{th}$ percentile and no one received SSI, WIC, SNAP, or Medicaid.

The explanatory variables consist of both the administrative data and the survey and frame data from the existing SIPP nonresponse adjustment. We estimate this model on respondents from the current panel and SIPP interview year in question. We then use the resulting decision rules for the CART leaves to assign nonrespondents (either Wave 1 nonrespondents or attriters for the subsequent waves) to the appropriate noninterview cell. We impose the restriction that each cell has at least 50 households (respondents or nonrespondents) to ensure the stability of the adjustments.

The CART model we have specified has several features beneficial for weighting the SIPP data. First, by using the aforementioned income and program participation dependent variable in the CART model, the CART procedure gives more weight to the administrative data that is most correlated with response outcomes. If, for example, the industry information from the business register has a low correlation with income and program participation (after controlling for other factors), then these variables will have low impacts on the CART cell definitions. Second, to handle non-linkages, we use the match indicators as explanatory variables in the CART model. If non-matched households have lower response rates, then the CART model can adjust for this. Thus, non-linkages with administrative data are not a problem for the CART model per se. The main issue is that for any noninterview adjustment cell of non-matched households, the missing conditionally at random assumption implicit in the weighting process is more questionable for these households than for other cells with more detailed administrative data. Third, the CART model is run separately for every SIPP year and panel. If the relationship between the survey and administrative data changes over time, then the CART model can account for that. For example, if more

people with zero income filed taxes curing the COVID-19 pandemic to receive COVID-related relief payments, then the CART model can account for changes in the distribution of IRS data over time.

For the noninterview adjustment, we use the administrative data attached to an SUID (the initial household identifier) from Wave 1. We do this for several reasons. For one, in the attrition adjustment, the survey data is also fixed to the Wave 1 values, so this allows us to use administrative data from the same time period as the survey data used as explanatory variables. Second, for households that split apart and one split off household does not respond, we may have more limited information to figure out the MAFID of these nonrespondent split-off household based on the procedure described in Section 2.3, given we do not have MAFID for the mover addresses.

Additionally, since the nonresponse adjustment in essence compares data between SIPP respondents and in-scope nonrespondents, the nonresponse adjustment for 2019 SIPP and beyond is affected by the unknown vacancy status issue described in Section 2.4. For SIPP respondents not matched to any administrative data, for example, their noninterview adjustment weight may assign these households too high of an intermediate weight because many vacant addresses would also have this characteristic. In future research, this could be addressed by predicting which households are vacant and adjusting the weight accordingly, such as what is described in U.S. Census Bureau (2021) and Keller et al. (2018). Using administrative data in the raking step where the targets are derived from known ACS occupied addresses should also help correct this issue, which is discussed in the following section.

## 3.3 Modification to Calibration Step

In this step, we also modify the second-stage raking procedure. This is potentially important because even if administrative data is used earlier in the algorithm, these adjustments to the weights may not be retained if only demographic population estimates are calibrated to in the final raking step. We add a step that calibrates SIPP weights to means of administrative data. Now the final raking step has three parts:

A. Spouse equalization
B. Calibration of SIPP's demographic data to estimates from the Population Estimates Program
C. Calibration to administrative data means

The raking algorithm repeats the three steps until convergence. The raking algorithm is performed on one month of SIPP data, so the raking procedure is repeated for each of the 12 calendar months. Our administrative data calibration step involves a set of discrete variables. We do not calibrate to all the

variables in our administrative dataset. We instead pick a combination of variables that are most related to income and program participation, as well as variables where nonresponse bias may be the highest. For example, we include moments on retirement status, given retired people may be both more likely to be at home when an interviewer comes, and have more time to complete the SIPP interview, which can be long. Unless noted otherwise, whether a household is matched to any of the main administrative data of IRS 1040, IRS 1099, SSR, and MAFARF is used as a calibration target as well in each step, so an indicator of missing data is used as a moment to calibrate the weights.

The administrative data moments are defined as

1. **Administrative data income**. Administrative data income is the sum of total money income in the IRS 1040 data, any retirement income from 1099-Rs not captured on the 1040 data, and any SSI income. Of households matched to any administrative data, households are divided into 10 bins based on percentiles of the income distribution. The income distribution is calculated from the ACS benchmark set of occupied housing units.
2. **Administrative data income interacted with demographics**. Households are divided up into four categories based on presence of children and presence of a married couple in the household. Within each of these four categories, households are divided into three income categories based on income percentiles.
3. **Foreign-Born Status**. Indicator of any foreign-born or non-citizens in the household, as indicated by information on the SSA Numident file.
4. **Medicaid Receipt**. Indicator of someone in household receiving Medicaid or CHIP.
5. **WIC Receipt**. Binary Indicator of WIC receipt.
6. **SNAP Receipt.** This categorical variable incorporates multiple parts. It has values for SNAP receipt as well as whether the household resides in a state where the Census Bureau has SNAP data coverage. Additionally, because presence of children is related to SNAP receipt, non-SNAP households are divided between households with and without children. This variable has the following levels:
    1. Someone in household receives SNAP.
    2. No one receives SNAP. Household has children and lives in a state with SNAP data coverage.
    3. No one receives SNAP. Household does not have children and lives in a state with SNAP data coverage.

4. No one receives SNAP. Household has children and does not live in a state with SNAP data coverage.
5. No one receives SNAP. Household does not have children and does not live in a state with SNAP data coverage.

7. **SSI and OASDI Disability Receipt.** A measure using the SSA program benefit data, defined primarily as
   1. Someone in household receives SSI disability benefits. No one receives OASDI disability benefits.
   2. No one receives SSI disability benefits. Someone receives OASDI disability benefits.
   3. Both SSI and OASDI disability recipients present in the household.
   4. No one receives SSI or OASDI disability benefits.
8. **SSI and OASDI Retirement Receipt.** A measure using the SSA program benefit data, defined primarily as
   1. Someone in household receives SSI aged benefits. No one receives OASDI retirement benefits.
   2. No one receives SSI aged benefits. Someone receives OASDI aged benefits.
   3. Both SSI aged and OASDI retirement recipients present in the household.
   4. No one receives SSI aged or OASDI retirement benefits.
9. **Employment Information.** A measure of employment and self-employment, defined as
   1. Someone in household matched to a 1040 record as primary or spouse filer with Schedule C or Schedule SE information.
   2. Someone in household matched to W-2 record, no one files Schedule C or SE.
   3. Household with no W-2, Schedule C, and Schedule SE data.
10. **Housing Structure Characteristics.** This variable captures various information about whether the housing structure is a single-family home based on information from the MAF, property assessed value from the third-party housing data, and itemization from 1040s. Itemization is used as a proxy for still having a sizable mortgage on the property.

This step helps provide more information on households not matched to IRS or SSA data. Bee et al. (2015) find that households not matched to IRS 1040 data tend to be in the tails of the income distribution. In this raking step, we separate out non-matched household living in apartments, who likely have lower income, from households in expensive single-family homes, who are likely higher income. With this information, this discrete variable is defined as follows:

1. Single-family home, not matched to third-party housing data or missing assessed valued in third-party housing data.
2. Single-family home, marked as not owner occupied based on information from the third-party housing data.
3. Single-family home, owner occupied, someone at this household itemizes. Then based on property assessed value, this set of households are subdivided into three bins of
    a. Property assessed value in bottom third of distribution.
    b. Property assessed value in middle third of distribution.
    c. Property assessed value in upper third of distribution.
4. Single-family home, owner occupied, no one itemizes. Then based on property assessed value, this set of households are subdivided into three bins of
    a. Property assessed value in bottom third of distribution.
    b. Property assessed value in middle third of distribution.
    c. Property assessed value in upper third of distribution.
5. Not a single-family home. Matched to an IRS 1040, IRS 1099, SSR, or MAFARF record.
6. Not a single-family home. Not matched to an IRS 1040, IRS 1099, SSR, or MAFARF record.

For each of these variables, we match the administrative data to SIPP respondents and ACS respondents and nonrespondents. The benchmark is calculated from the set of ACS respondents and nonrespondents, and the SIPP weights are calibrated to that ACS-Adrec estimate.

A detailed description of our procedures for calibration to administrative data means is as follows:

1. For each of the discrete administrative data variables $k$ described above, let $y_{i,t,s}^{k,c}$ be an indicator that equals one if household $i$ in year $t$ of survey $s$ (which equals $a$ for ACS and $p$ for SIPP) is in cell $c$ for administrative data variable $k$.
2. Calculate the weighted percentage of respondents and in-scope nonrespondents in each of these $C^k$ cells for administrative data variable $k$, using the ACS benchmark set of observation with the base weight times the CAPI subsampling factor. This is the target to calibrate to, which we denote by $\hat{\pi}_t^{k,c}$.
3. For the raking procedure, add these above targets to the calibration step. Note that the above targets are household level, while the raking step in question is for constructing person-month-level weights. In this step, we divided the current person's weight by the number of household members, given each household member will have the same value of the household-level

target. The number of household members is the number of people in that month sharing the same household identifier.[5] This is very similar to the Entropy-Balance Weighting approach used in Rothbaum et al. (2021) for the 2020 ACS, that did a similar scaling when calibrating to administrative data-derived targets in the final calibration step. This procedure of dividing the weights by the number of household members helps ensure that each person contributes to satisfying the moments from linked decennial census, administrative, and third-party data, which should reduce the variability of weights among household members. This can be particularly important for person-level statistics, such as poverty. Given this, the exact mathematical procedure is as follows:

a. Let $n_{i,t,m,p}^{HH}$ denote the number of household members in household $i$ in year $t$ of month $m$ of the SIPP survey, and $\widetilde{w}_{i,j,t,m,p}$ be person $j$'s current weight in the raking algorithm for month $m$.
b. Let the new weight be given by the following equation:

$$\widetilde{w}_{i,j,t,m,p} * \hat{\pi}_t^{k,c} * \left( \sum_{i,j \in R} \frac{\widetilde{w}_{i,j,t,m,p}}{n_{i,t,m,p}^{HH}} \right) / \left( \sum_{i,j \in R} \frac{y_{i,t,p}^{k,c} \widetilde{w}_{i,j,t,m,p}}{n_{i,t,m,p}^{HH}} \right)$$

c. This equation is the current person's weight multiplied by the target percentage of households in raking category $c$ divided by the current calculated percentage of household in that category, using the modified version of the weights.
d. As an additional step that is likely not necessary, we also adjust the total of the weights to equal the population total after this adjustment. We add this because this ratio adjustment may not preserve the population total of the weights near the beginning of the raking algorithm. However, this ratio adjustment will get close to one near convergence, so this step is likely unnecessary.
e. Do this raking step for each of the $K$ administrative data variables.

We perform this calibration for each set of replicate weights for constructing the standard errors. One complication is that there are 240 replicates in SIPP but only 80 for ACS. For this research, given 240 is divisible by 80, we repeat the ACS replicates when looping through the SIPP replicates. For example, for SIPP replicate number 82, we use the 2nd replicate base weight from ACS. While this is not proper statistical methodology, the administrative data means calculated from the ACS have small confidence

[5] On the public use data, these would be individuals sharing the same ssuid and shhadid.

intervals given the very large sample size of the ACS. Future research can explore how best to incorporate the uncertainty in the ACS-based administrative data estimates into SIPP's weighting process.

## 3.4 Comparison to Prior Administrative Data Weighting Papers

There has been prior work at the Census Bureau exploring adding administrative data to the weighting process for household surveys. Eggleston and Westra (2020) analyze adding administrative data to SIPP's weighting algorithm for 2014 SIPP Wave 1. In this paper, we add much more administrative data to the weighting model. In particular, this paper uses program benefit data on SSA benefits, Medicaid, SNAP, WIC, and TANF, which were not used in the prior paper. Additionally, Eggleston and Westra (2020) only modify the household noninterview adjustment for wave 1, while this paper also modifies the raking step and household noninterview adjustment for waves 2+ with administrative data. Section 4.3 presents results to show how modifying the raking step changes SIPP estimates compared to only modifying the noninterview adjustment.

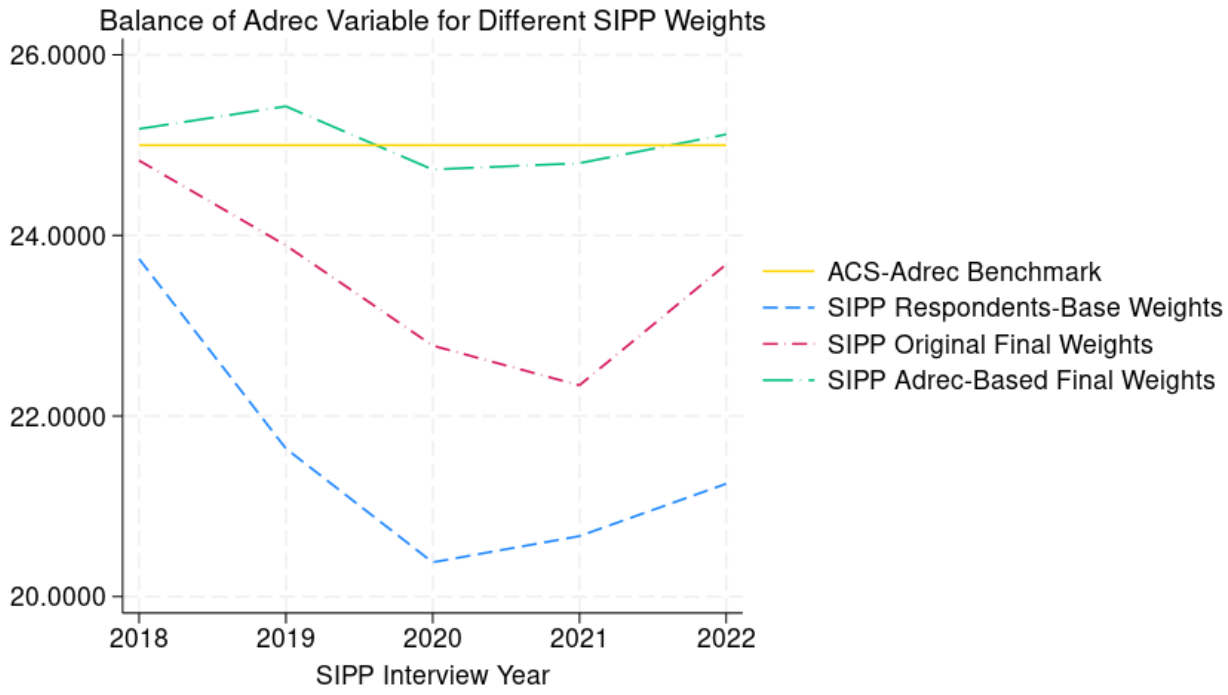
Two of the other administrative data weighting research papers mentioned in the introduction, Rothbaum et al. (2021) on the ACS and Rothbaum and Ash (2024) on the Consumer Expenditure Survey, have used an Entropy Balance Weighting algorithm, which employs an optimization algorithm that calibrates the weighting to administrative data moments and demographic estimates from the Population Estimates Program subject to minimizing a loss function that penalizes the change in someone's weight between their initial weight and final weight. The benefits of employing such an algorithm over raking is left to future research. Deville and Särndal (1992) show the theoretical equivalence of raking and optimization-based approaches under certain conditions, and Slud and Morris (2022) discuss how these approaches compare under more typical survey conditions. The main reason we chose to use a raking approach was to use the pre-existing SIPP weighting code as much as possible. For any potential future implementation of this research in SIPP production, the raking change should be easier to implement than an entropy balance weighting algorithm, the latter of which is more of a complete replacement of survey weighting procedures rather than a modification. Thus, in terms of implementation, incorporating administrative data via the raking step may yield benefits from an operational perspective.

# 4. Results

## 4.1 Nonresponse Bias in Administrative Data

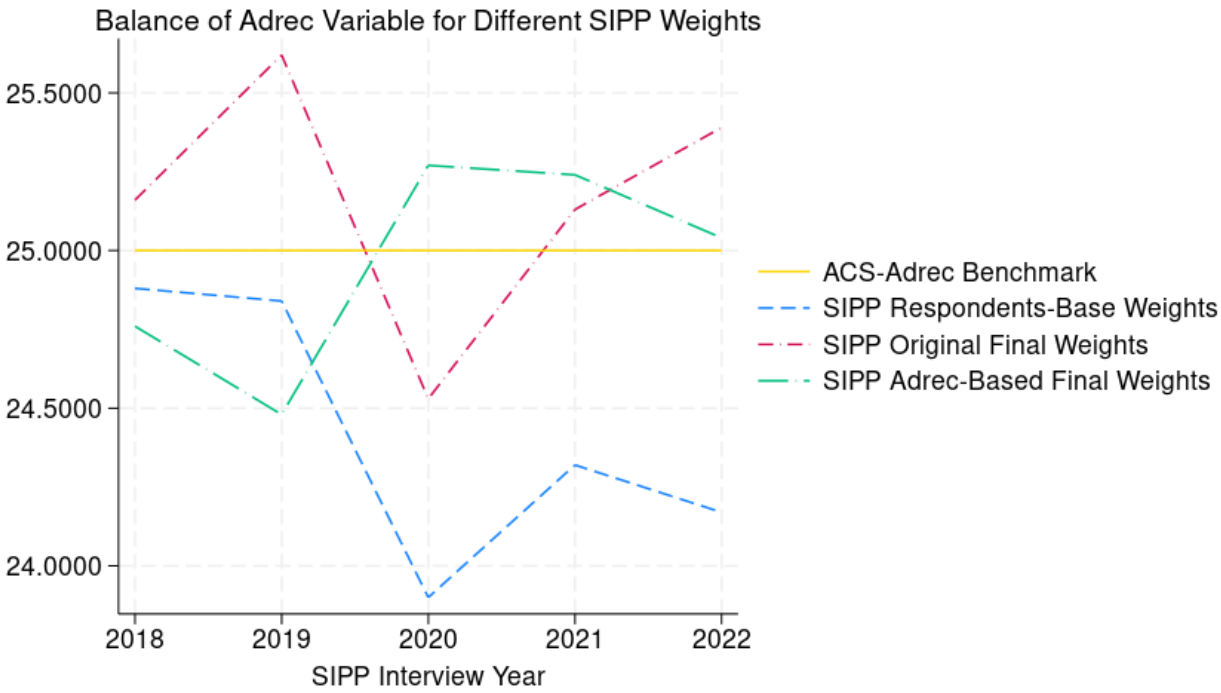
In this section, we present results on how representative the SIPP sample is using the administrative data described in Section 2. First, we calculate means of administrative data variables from the ACS benchmark set of observations, which includes respondent and nonrespondent households in occupied housing units. Next, we estimate the same administrative data means from all SIPP respondents using a modified version of the base weights. We take a household's base weight from Wave 1 and then multiply it by the moving factor and the overlapping panel factor. This allows us to pool all the panels in a calendar year together in a similar manner as what is done in the SIPP weighting process. The goal of presenting this mean with modified base weights is to show how representative the SIPP sample is before weighting adjustment. Finally, we estimate administrative data means using both the original final weights and the new adrec-based final weights. To create household-level estimates from these person-month weights, we divide the final weights by the number of household members in that month, similar to what is described in Section 3.3 for the raking step. Using every household member's weight in these calculation helps show how balanced the entire sample is, rather than if we had picked just one person's weight like the household reference person.

Figure 3: Percent with Administrative Data Income Below 25th Percentile, Households with Children



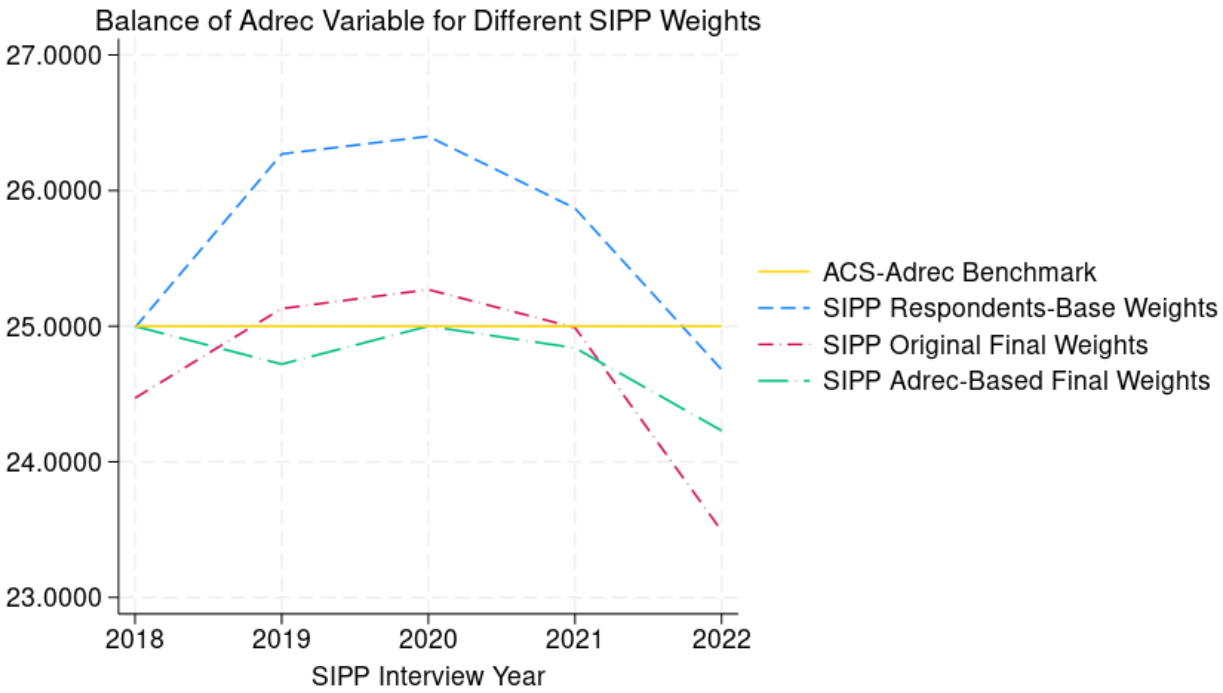

Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data.

Figure 4: Percent with Administrative Data Income Between 25th and 50th Percentile, Households with Children



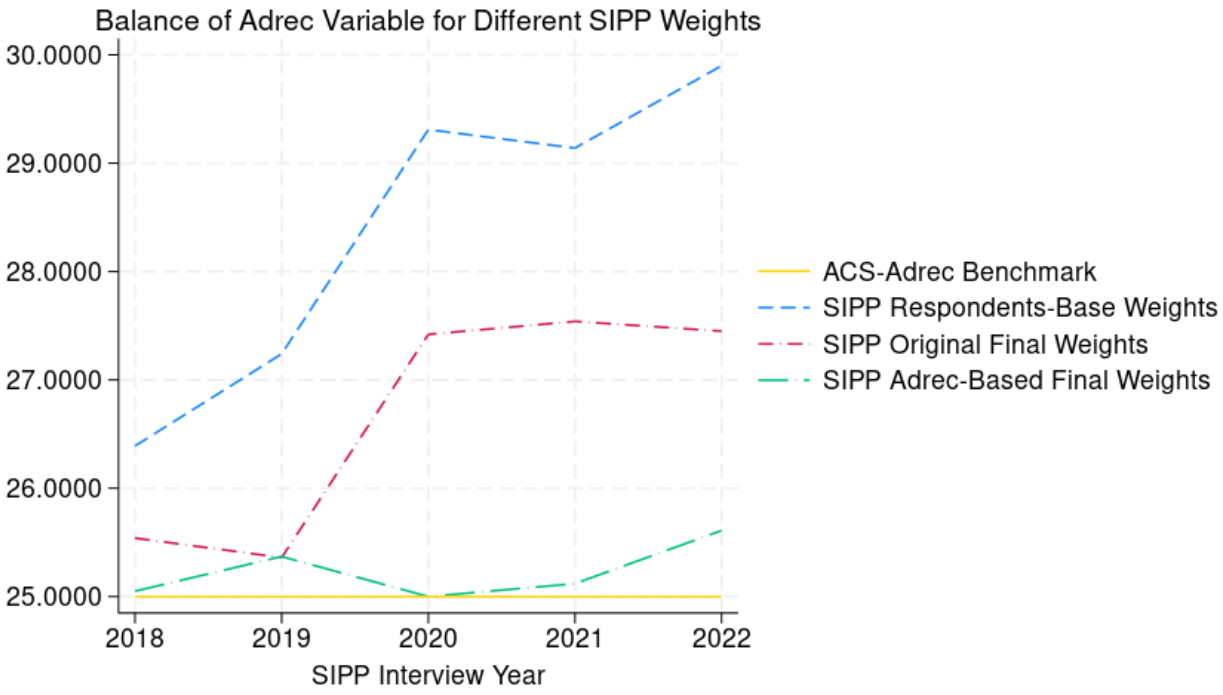

Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data.

Figure 5: Percent with Administrative Data Income Between 50th and 75th Percentile, Households with Children



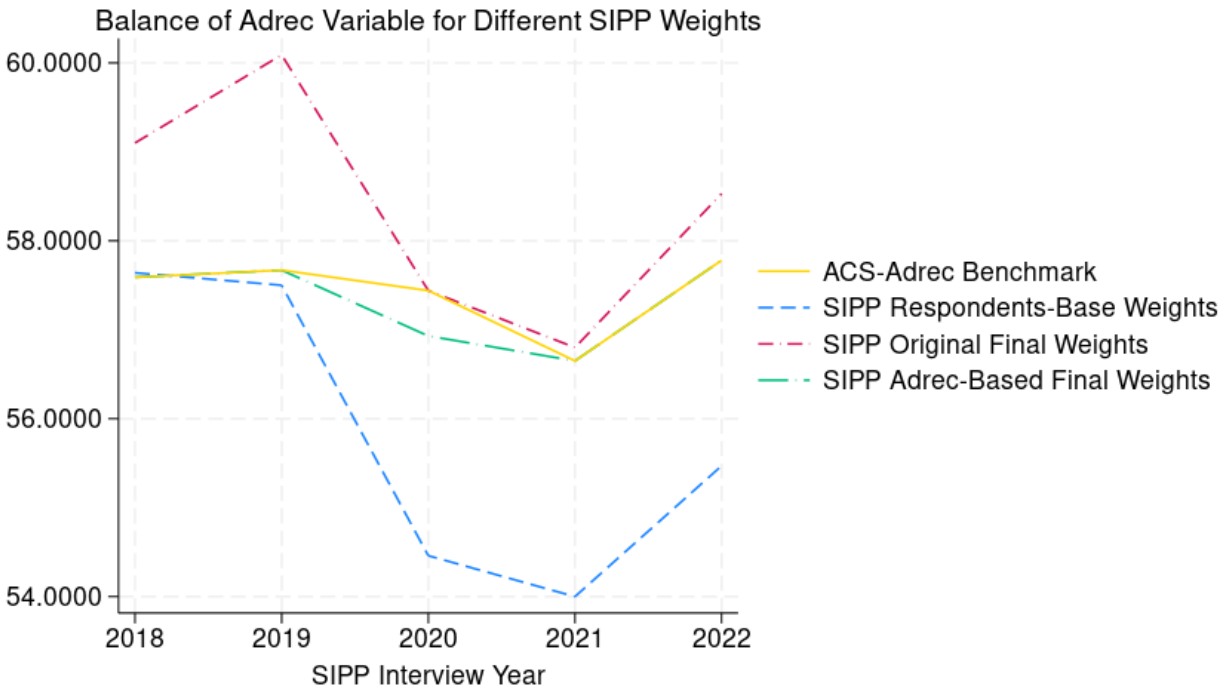

Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data.

Figure 6: Percent with Administrative Data Income Above 75th Percentile, Households with Children



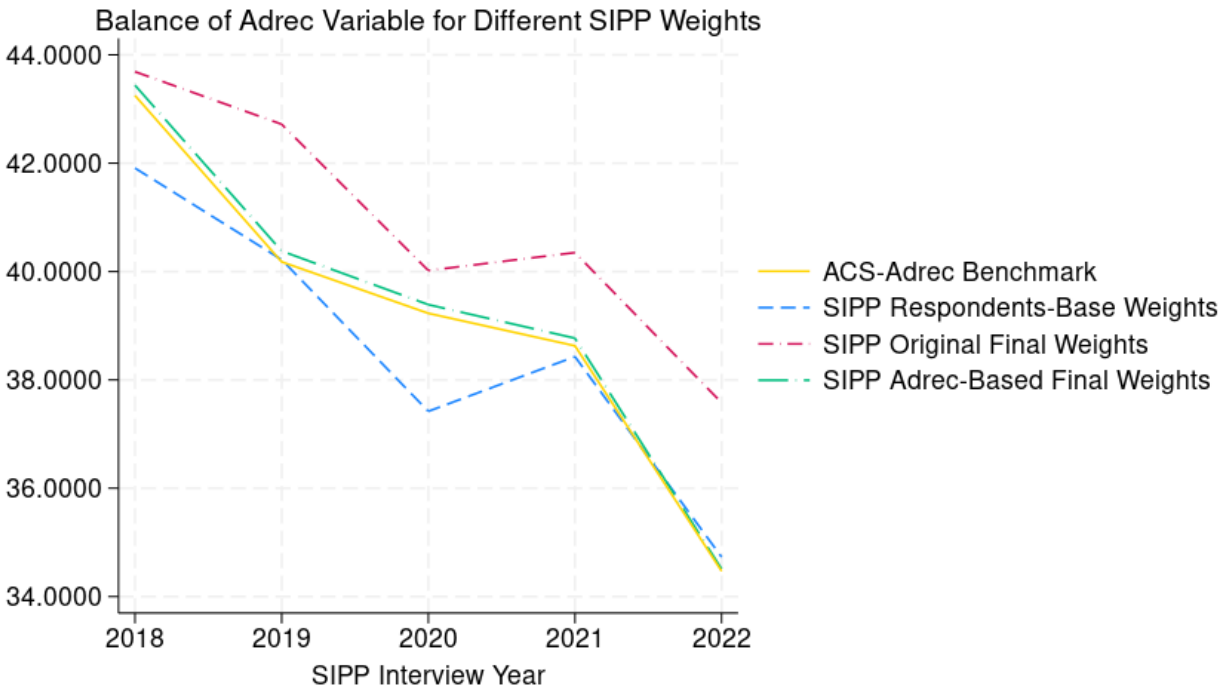

Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data.

Figure 7: Medicaid Receipt Rates, Households with Children




Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data.

Figure 8: SNAP Receipt Rates, Households with Children




Source: ACS, SIPP, and administrative data. Estimates restricted to household with children under 18, as measured in the administrative data. Restricted to states with available SNAP administrative data in that year.

Table A.1 shows the full set of balance results with respect to various administrative data variables. To help summarize the results visually, we create graphs in Figures 3 through 8 that focus on households with children in the administrative data. We do this to condition on a set of households in a similar life stage. This helps provide a picture of how bias with respect to income and program participates varies for different sets of weights, after controlling for some demographic variables.

Figures 3 through 6 examine income. The income percentiles are defined from the ACS benchmark set of households. Because of this, the ACS-Adrec benchmark is 25 percent in all these graphs by construction. Before weighting adjustments, the graphs show SIPP respondents having higher income than the general population. SIPP respondents are more likely to have income above the 75$^{th}$

percentile, and for some SIPP years they are also more likely to be between the 50th and 75th percentile.[6] SIPP respondents are less likely to be under the 25th percentile.

How well existing SIPP weighting procedures address these income differences is complicated. For both the percentage of households under the 25th percentile and over the 75th percentile in 2018 through 2022 SIPP, the production weighing procedures reduce the difference between SIPP respondent and the benchmark, although some differences remain. For these measures in 2020-2022 SIPP, the means with the adrec-based weights are closer to the benchmark than the means with production weights. The results are more muddled in the middle of the income distribution. For 2022 SIPP and the percent of households between the 25th and 50th percentile, for example, the base weighted mean is lower than the benchmark, while the mean with the production weights is over the benchmark.

Medicaid receipt also shows sign reversal in some years. For 2022 SIPP, respondents are less likely to be receiving Medicaid according to the base-weighted measured. However, the mean with the production final weights is above the ACS benchmark. For SNAP, the estimated base-weighted mean for SIPP is not significantly different from the estimated ACS benchmark in 2019, 2021, and 2022. However, in these years, the production weights appear to create a bias; the means with SIPP production weights are higher than the ACS benchmark.

All together, these graphs of households with children suggest that the magnitude and sign of nonresponse bias in SIPP may not be uniform across the joint income and program participation distribution. In 2022 SIPP, for example, the production weights appear to generate too few low-income households (measured by percentage with income below 25th percentile) and too many households receiving Medicaid, which is surprising given the relationship between income and Medicaid eligibility. This suggests that when adding administrative data to SIPP weighting, there may also be changes in estimates that are not uniform across the joint income and program participation distribution and other variables correlated with these measures.

[6] For income between the 50th and 75th percentiles, the estimate difference is insignificant for 2018 and 2022.

## 4.2 Weight Distribution Statistics

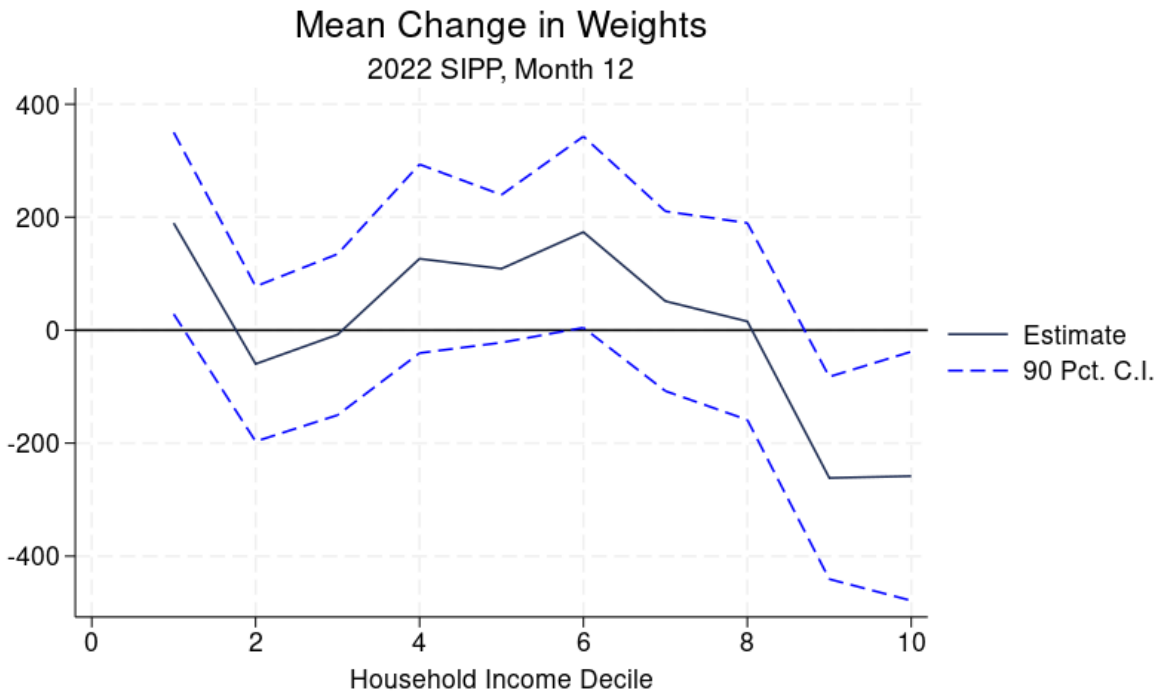
Table 5: Change in Weight Distribution, 2022 SIPP Month 12

| Statistic | Production Weight | Adrec-Based Weight | P-Value Difference |
|---|---|---|---|
| Mean | 8,053 | 8,053 | N.A. |
| S.D. | 4,578 | 4,666 | 0.2974 |
| Skewness | 2.825 | 2.289 | 0.3832 |
| Kurtosis | 21.25 | 13.78 | 0.3698 |
| p5 | 3,353 | 3,040 | 0.455 |
| p10 | 3,883 | 3,655 | 0.3144 |
| p25 | 5,072 | 4,933 | 0.4545 |
| p50 | 7,004 | 6,933 | 0.2757 |
| p75 | 9,703 | 9,908 | 0.6799 |
| p90 | 13,230 | 13,700 | 0.0259 |
| p95 | 16,220 | 16,590 | 0.5488 |

Source: 2022 SIPP, Month 12 and administrative data. Table presents summary statistics on final weights between the production version and the adrec-based weight version.

To dive further into the effect of adding administrative data to SIPP's weighting processes, we present summary statistics on the distribution of the weights themselves to provide more information. First, in Table 5, we present summary statistics on the final weights for month 12 of 2022 SIPP (reference month December 2021). Overall, there is no significant change in these distribution statistics. The one significant change is for the 90$^{th}$ percentile of the distribution of weight values, which increased from 13,230 to 13,700.

Figure 9: Mean Change in Weights by Household Income, 2022 SIPP Month 12

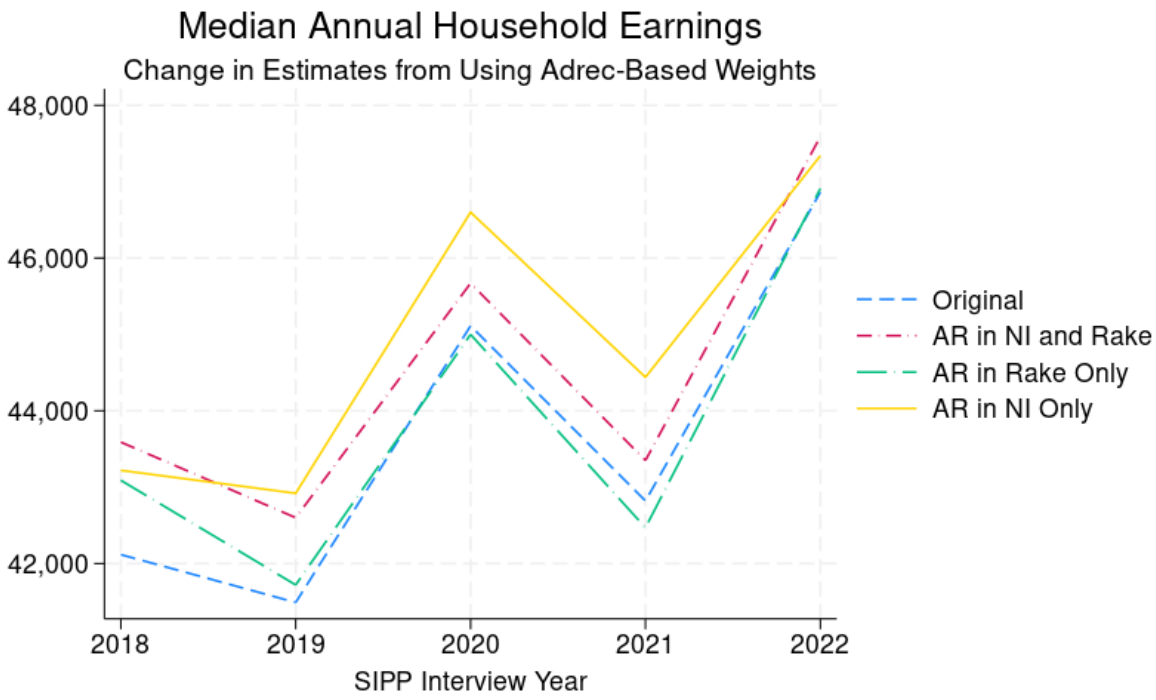



Source 2022 SIPP month 12 and administrative data. Weight values are winsorized to the 1st and 99th percentile.

In the prior section, we presented evidence that the change in weight values may not be uniform across the income distribution. We investigate this more fully in Figure 9, which presents weight statistics once again for 2022 SIPP month 12. The figure shows the average change in weights by SIPP household income (survey data values, not administrative data income values). This shows that the 1st decile has higher mean weight values with adrec-based weights, and the 9th and 10th deciles have lower mean weight values with adrec-based weights. However, the changes are not monotonic through the income distribution. For example, the 2nd decile of income has an average change of -60, while the 6th decile has an average change of 174. This provides further evidence that the effects of adding administrative data to the SIPP weighting process is complicated. While the decrease in the weight values for the highest quintile matches a story of SIPP respondents having higher socio-economic status than the general population, the point estimate of the mean weight change for some quintiles in the middle of the income distribution is actually higher than the weight change for households lower in the income distribution.

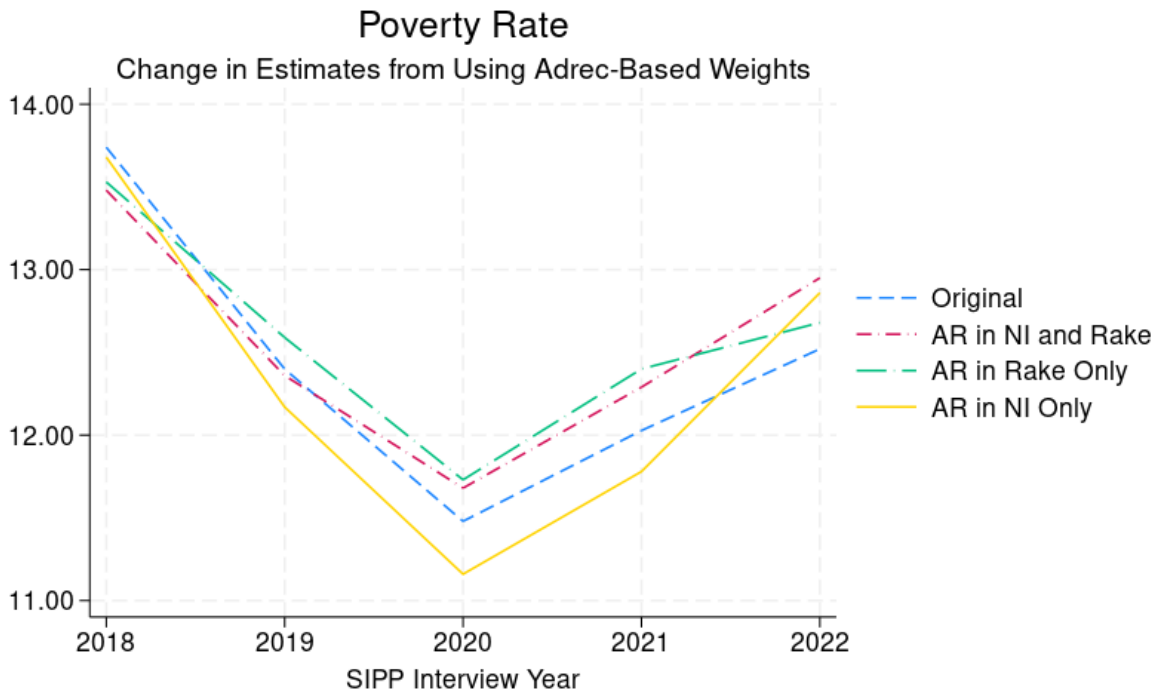
## 4.3 Change in Survey Estimates

Figure 10: Change in Estimates for Median Annual Household Income



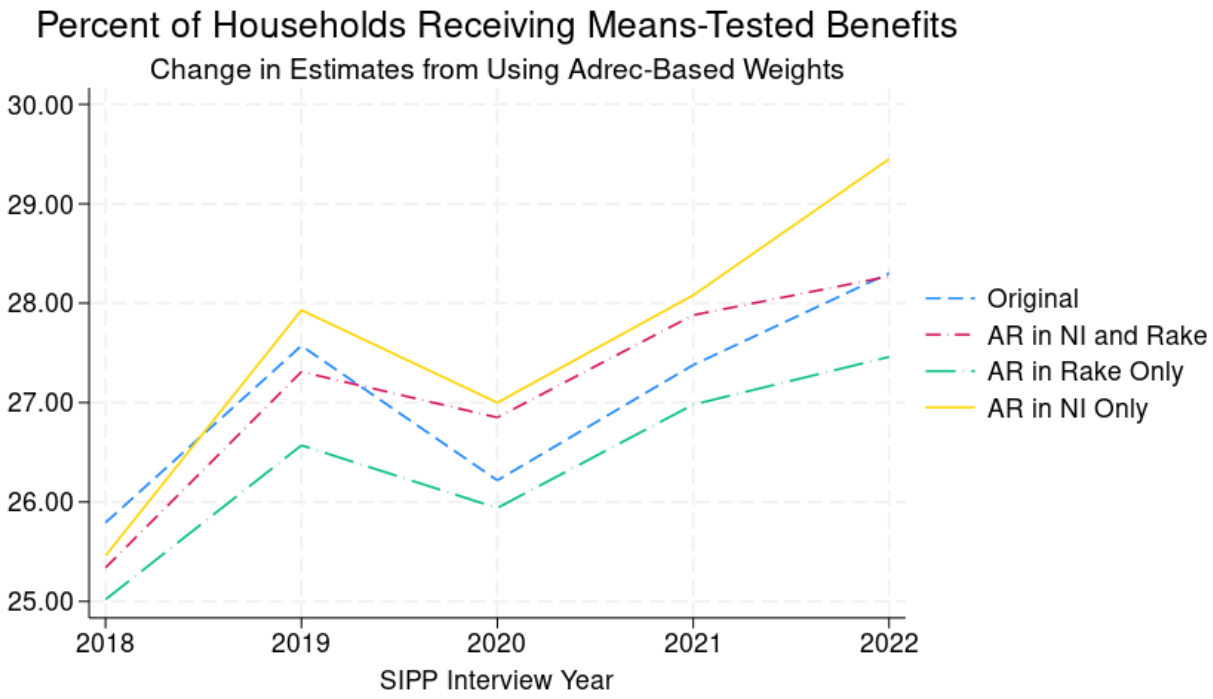

Source: SIPP and administrative data.

Figure 11: Change in Estimates for Poverty



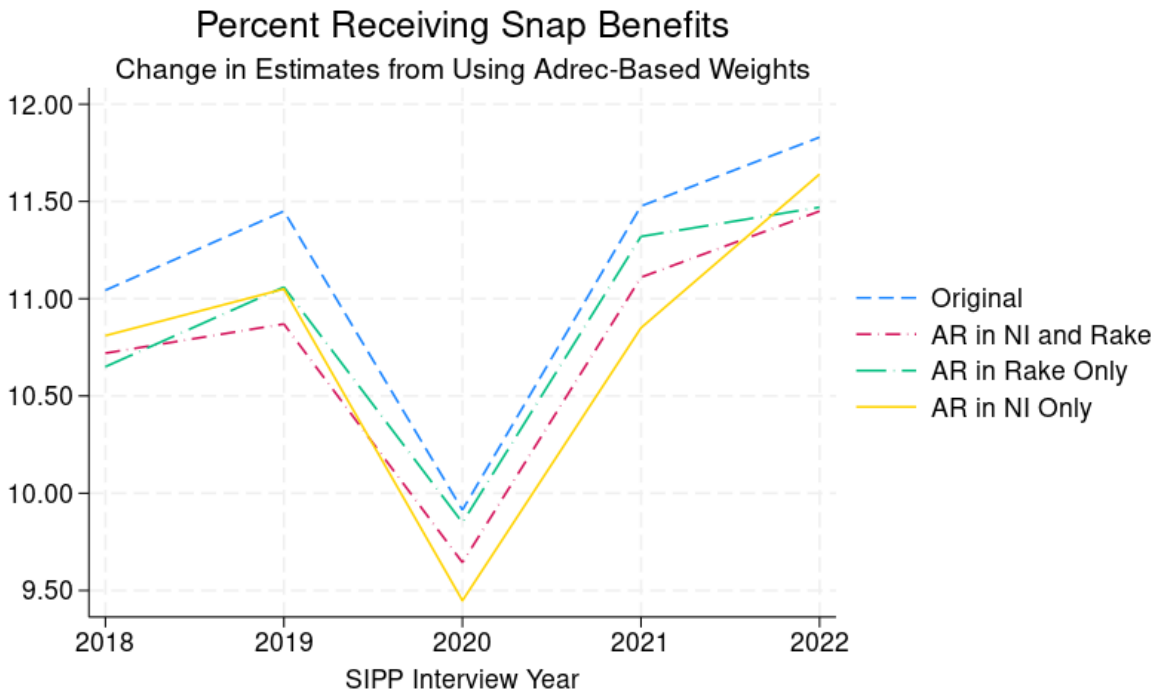

Source: SIPP and administrative data.

Figure 11: Change in Estimates for Percent of Households Receiving Means-Tested Benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC)



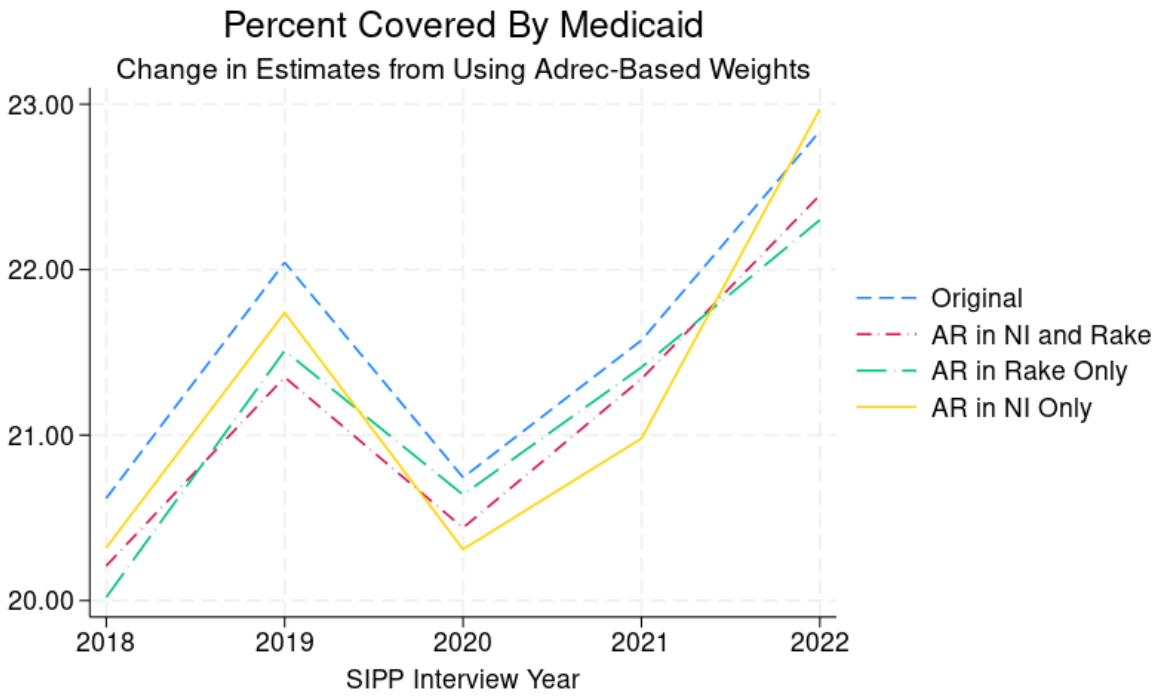

Source: SIPP and administrative data.

Figure 13: Change in Estimates for Percent Receiving SNAP Benefits



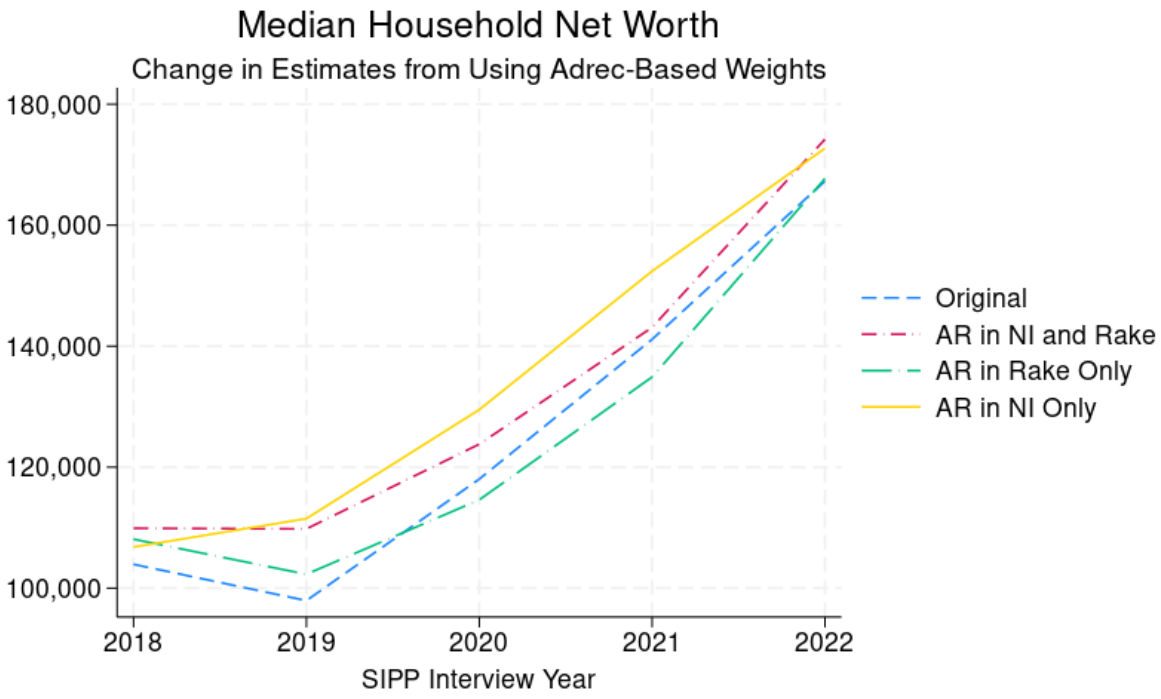

Source: SIPP and administrative data.

Figure 14: Change in Estimates for Percent Receiving Medicaid




Source: SIPP and administrative data.

Figure 15: Change in Estimates for Median Net Worth




Source: SIPP and administrative data.

In this section, we present results on how adding administrative data to SIPP's weighting process changes survey estimates. In existing SIPP data review and quality monitoring practices, there are a set of 27 statistics labeled "key estimates" from a variety of topic areas. They represent some of the more important or widely used estimates from the SIPP survey data. Table A.2 presents the results for all the key estimates, and Figures 10 through 15 take the results from some of the variables presented in Table A.2 and present them graphically. The table and figures present three different sets of estimates with administrative data-based weights, corresponding to

1. Adding administrative data to both the household noninterview adjustment and the second stage calibration (raking step),
2. Adding administrative data just to the household noninterview adjustment, and
3. Adding administrative data just to the second-stage calibration.

Adding administrative data to both the noninterview adjustment and the calibration step is our primary specification, and was the specification shown in Sections 4.1 and 4.2. We have the other specifications to understand how important adding the administrative data is to each part of the weighting process.

For example, if estimates in #1 and #3 are very similar, then for any future changes to SIPP's weighting process, modifying the calibration step may be the only modification that is needed.

Overall, the sign and magnitude of the change in estimates differs by both year and variable. First, examine the change in estimates for 2022 SIPP. Using administrative data in survey weights results in higher estimates of the poverty rate. With respect to this variable, the adrec-based weights result in lower estimates of economic wellbeing. However, these same weights result in lower estimates of person-level SNAP receipt, and higher estimates of median household wealth. Additionally, the household-level measure of receiving any means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) is unchanged. Thus, the direction of the changes is variable-specific and does not move in the same direction one would expect given the positive correlation between poverty status and receipt of government benefits. This further supports the results from section 4.2 that the effects of the administrative data-based weight are not monotonic across the income distribution.

The direction of the changes also varies by year for some variables. For example, administrative data-based weights results in a higher estimate of median household income for 2018 SIPP but a lower estimate for 2021 SIPP. In terms of the different weight specifications, the different weighting models do often result in different estimates, although it is debatable how large or meaningful these differences are. The rates of household-level receipt of means-tested benefits does show differences by the weighting choice. In 2022, using administrative data in both the noninterview adjustment and the calibration step results in an estimate of 28.27% of household receiving such benefit. With administration data in only raking, the estimate is 27.46%, lower than the production estimate. Using administrative data only in the noninterview adjustment results in an estimate of 29.45%, higher than the production estimate.

# 5. Discussion and Conclusion

In this paper, we examine the effects of adding administrative data to SIPP's weighting algorithm for survey years 2018 to 2022. These administrative data include income data from IRS and program benefit data on SSA benefits, Medicaid, SNAP, WIC, and TANF. Adding administrative data to the weighting process does result in a change to SIPP estimates. The sign and magnitude vary by both year and variable. We attribute these differential effects to the change in weight values not being monotonic throughout the income distribution.

There is no notable spike in nonresponse bias in 2020 SIPP, which was subject to COVID-19 related interviewing restrictions. This is unlike the ACS and CPS ASEC, which experience a large increase

in nonresponse bias in that interviewing year. Potential reasons for this could be SIPP having lower response rates in preceding years to 2020, or the existing SIPP weighting procedures making use of economic measures like the high-poverty stratum flag in the weighting model.

The new weighting approach is designed to minimize one specific source of bias affecting survey: nonresponse bias. From a total survey error perspective (Groves and Lyberg, 2010) the accuracy of statistics is affected by other form of bias as well, such as coverage error and measurement error. These weighting approaches do nothing to correct for any measurement error in SIPP, such as SNAP beneficiaries failing to report these benefits on surveys (Celhay et al. 2024). If nonresponse bias and measurement error bias for a variable have different signs, it is possible that correcting nonresponse bias could in fact increase overall bias in an estimate. Using SNAP once again as an example, results from prior literature indicate respondents underreport receipt of SNAP benefits, so measurement error bias for this variable is a downward bias. Our new weighting approach results in a lower estimate of SNAP receipt, suggesting than nonresponse bias for this variable in SIPP is a slight upward bias. Given the measurement error bias is much larger (Celhay et al. 2024 find a false negative rate of 18% for SIPP), our lower estimate of SNAP receipt is likely a more biased estimate than the production estimate, once all sources of error are accounted for. This is happening even though we are using SNAP administrative data in our weighting model.

Overall, our results suggest that the sign and magnitude of nonresponse bias in SIPP may be hard to predict ahead of time. Given variability in sample sizes and interviewer variability, the characteristics of who responds to SIPP seems to change over time. How well the existing weighting processes correct for these differences is complicated, given the existing weighting process has multiple inputs. Additionally, the nature of nonresponse bias in SIPP may be different in the future with the "SIPP Seamless" initiative, which incorporates a large-scale questionnaire redesign which is expected to yield fewer questions, a switch to year-round interviewing to employ interviewers continuously instead of hiring interviewers on a temporary basis each year, multiple interviews per year of varying length, and the introduction of an Internet Self-Response mode option. In future work, administrative data could be used to investigate how the characteristics of SIPP respondents change under the SIPP Seamless design. Administrative data-based weights could help reduce the effects of nonresponse changing over time due to either design changes with the survey, changes in the availability of interviewers, or changes in the general willingness of individuals to respond to government surveys.

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# Appendix

Table A.1: Balance of Administrative Data Variables

| Adrec. Variable | SIPP Year | ACS-Adrec Benchmark | S.E. | SIPP Respondents-Base Weights | S.E. | Adrec-Based Final Weight | S.E. | Original Final Weight | S.E. |
|---|---|---|---|---|---|---|---|---|---|
| Income Greater Than 75th Percentile | 2018 | 25.00 | 0.0812 | 25.08 | 0.2855 | 24.85 | 0.2801 | 24.91 | 0.1347 |
| Income Greater Than 75th Percentile | 2019 | 25.00 | 0.0445 | 25.16 | 0.3419 | 24.93 | 0.3517 | 24.91 | 0.13 |
| Income Greater Than 75th Percentile | 2020 | 25.00 | 0.0458 | 26.39 | 0.2966 | 24.96 | 0.3097 | 26.19 | 0.1207 |
| Income Greater Than 75th Percentile | 2021 | 25.00 | 0.0958 | 26.48 | 0.3011 | 24.95 | 0.29 | 26.47 | 0.147 |
| Income Greater Than 75th Percentile | 2022 | 25.00 | 0.0455 | 26.71 | 0.4097 | 24.92 | 0.4358 | 25.85 | 0.1759 |
| Income Less Than 25th Percentile | 2018 | 25.00 | 0.0864 | 24.75 | 0.2994 | 25.00 | 0.2732 | 25.06 | 0.1352 |
| Income Less Than 25th Percentile | 2019 | 25.00 | 0.0400 | 23.53 | 0.3197 | 25.06 | 0.3447 | 24.65 | 0.1388 |
| Income Less Than 25th Percentile | 2020 | 25.00 | 0.0482 | 23.62 | 0.2980 | 24.93 | 0.32 | 24.09 | 0.1292 |
| Income Less Than 25th Percentile | 2021 | 25.00 | 0.0847 | 23.97 | 0.2910 | 24.94 | 0.3325 | 23.81 | 0.1529 |
| Income Less Than 25th Percentile | 2022 | 25.00 | 0.0433 | 24.40 | 0.3376 | 24.81 | 0.3653 | 24.86 | 0.1475 |
| Income Between 25th and 50th Percentile | 2018 | 25.00 | 0.0929 | 25.29 | 0.2877 | 25.00 | 0.2929 | 25.27 | 0.1372 |
| Income Between 25th and 50th Percentile | 2019 | 25.00 | 0.0369 | 25.85 | 0.3044 | 24.94 | 0.3682 | 25.54 | 0.1306 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Income Between 25th and 50th Percentile | 2020 | 25.00 | 0.0460 | 24.66 | 0.2743 | 25.07 | 0.2642 | 24.51 | 0.1296 |
| Income Between 25th and 50th Percentile | 2021 | 25.00 | 0.0779 | 24.91 | 0.3197 | 25.06 | 0.3475 | 25.01 | 0.1386 |
| Income Between 25th and 50th Percentile | 2022 | 25.00 | 0.0481 | 24.15 | 0.3813 | 25.19 | 0.4689 | 24.88 | 0.1522 |
| Income Between 50th and 75th Percentile | 2018 | 25.00 | 0.0844 | 24.87 | 0.3040 | 25.15 | 0.2944 | 24.76 | 0.1354 |
| Income Between 50th and 75th Percentile | 2019 | 25.00 | 0.0334 | 25.46 | 0.3040 | 25.07 | 0.3416 | 24.90 | 0.1262 |
| Income Between 50th and 75th Percentile | 2020 | 25.00 | 0.0452 | 25.34 | 0.3147 | 25.04 | 0.315 | 25.22 | 0.125 |
| Income Between 50th and 75th Percentile | 2021 | 25.00 | 0.0833 | 24.64 | 0.3032 | 25.05 | 0.3404 | 24.71 | 0.1383 |
| Income Between 50th and 75th Percentile | 2022 | 25.00 | 0.0435 | 24.73 | 0.3588 | 25.08 | 0.3874 | 24.40 | 0.175 |
| Any Over 60 | 2018 | 41.24 | 0.1020 | 46.00 | 0.3258 | 41.72 | 0.2405 | 42.67 | 0.1928 |
| Any Over 60 | 2019 | 42.23 | 0.0407 | 52.40 | 0.3793 | 42.50 | 0.3759 | 44.00 | 0.2502 |
| Any Over 60 | 2020 | 42.94 | 0.0501 | 53.65 | 0.3810 | 43.80 | 0.3308 | 45.59 | 0.2485 |
| Any Over 60 | 2021 | 43.38 | 0.0956 | 54.39 | 0.3542 | 44.35 | 0.2889 | 46.49 | 0.2028 |
| Any Over 60 | 2022 | 43.41 | 0.0484 | 53.74 | 0.4395 | 43.70 | 0.3377 | 46.15 | 0.2304 |
| Any Children Under 5 | 2018 | 19.21 | 0.0787 | 18.31 | 0.2713 | 19.65 | 0.2488 | 19.40 | 0.2028 |
| Any Children Under 5 | 2019 | 18.41 | 0.0349 | 16.00 | 0.2833 | 18.85 | 0.2839 | 18.66 | 0.2419 |
| Any Children Under 5 | 2020 | 18.18 | 0.0427 | 15.89 | 0.2402 | 18.83 | 0.2724 | 18.78 | 0.2151 |
| Any Children Under 5 | 2021 | 17.51 | 0.0757 | 15.27 | 0.2406 | 18.04 | 0.2478 | 18.17 | 0.1869 |
| Any Children Under 5 | 2022 | 16.88 | 0.0392 | 14.12 | 0.2891 | 16.67 | 0.2947 | 16.46 | 0.2415 |
| Anyone with 4+ W-2s | 2018 | 5.77 | 0.0484 | 5.68 | 0.1570 | 5.84 | 0.1637 | 6.03 | 0.1568 |
| Anyone with 4+ W-2s | 2019 | 6.01 | 0.0255 | 5.77 | 0.2074 | 6.05 | 0.2423 | 6.53 | 0.2169 |
| Anyone with 4+ W-2s | 2020 | 6.05 | 0.0234 | 5.74 | 0.1904 | 5.90 | 0.2134 | 6.35 | 0.2001 |
| Anyone with 4+ W-2s | 2021 | 4.76 | 0.0451 | 4.63 | 0.1546 | 4.70 | 0.197 | 5.39 | 0.1766 |
| Anyone with 4+ W-2s | 2022 | 5.91 | 0.0258 | 5.77 | 0.2056 | 5.83 | 0.2315 | 6.33 | 0.235 |
| Any Asian | 2018 | 6.40 | 0.0456 | 6.50 | 0.1923 | 6.57 | 0.1645 | 6.69 | 0.159 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Any Asian | 2019 | 6.34 | 0.0260 | 5.99 | 0.2000 | 6.31 | 0.1959 | 6.61 | 0.1889 |
| Any Asian | 2020 | 6.38 | 0.0212 | 6.66 | 0.1934 | 6.62 | 0.1486 | 7.12 | 0.1416 |
| Any Asian | 2021 | 6.39 | 0.0519 | 6.42 | 0.1886 | 6.31 | 0.168 | 6.75 | 0.1616 |
| Any Asian | 2022 | 6.40 | 0.0248 | 7.18 | 0.2053 | 6.64 | 0.1884 | 6.83 | 0.1868 |
| Any Black | 2018 | 14.64 | 0.0737 | 13.62 | 0.2139 | 15.33 | 0.1787 | 15.40 | 0.1693 |
| Any Black | 2019 | 14.72 | 0.0368 | 12.24 | 0.2366 | 15.35 | 0.2341 | 15.61 | 0.2446 |
| Any Black | 2020 | 14.76 | 0.0310 | 12.72 | 0.2221 | 15.55 | 0.2481 | 15.91 | 0.2435 |
| Any Black | 2021 | 14.79 | 0.0795 | 12.44 | 0.2138 | 15.09 | 0.2223 | 15.61 | 0.2307 |
| Any Black | 2022 | 14.64 | 0.0346 | 12.10 | 0.2766 | 15.19 | 0.2984 | 15.47 | 0.2846 |
| Any Children Under 5 | 2018 | 43.27 | 0.0957 | 41.21 | 0.3247 | 43.27 | 0.2742 | 42.90 | 0.095 |
| Any Children Under 5 | 2019 | 42.70 | 0.0423 | 37.32 | 0.3409 | 42.70 | 0.3104 | 41.92 | 0.042 |
| Any Children Under 5 | 2020 | 42.61 | 0.0609 | 36.86 | 0.3344 | 42.61 | 0.3282 | 41.72 | 0.0612 |
| Any Children Under 5 | 2021 | 41.37 | 0.0983 | 35.78 | 0.3366 | 41.37 | 0.3067 | 40.56 | 0.0959 |
| Any Children Under 5 | 2022 | 40.95 | 0.0524 | 36.26 | 0.4317 | 40.95 | 0.3848 | 40.24 | 0.0517 |
| Any Deferred Earnings | 2018 | 42.97 | 0.1001 | 43.90 | 0.3619 | 43.53 | 0.342 | 43.78 | 0.2924 |
| Any Deferred Earnings | 2019 | 43.33 | 0.0489 | 46.10 | 0.3920 | 44.44 | 0.4497 | 46.04 | 0.3917 |
| Any Deferred Earnings | 2020 | 44.82 | 0.0453 | 47.69 | 0.3851 | 45.60 | 0.4009 | 47.58 | 0.3311 |
| Any Deferred Earnings | 2021 | 44.24 | 0.0999 | 47.55 | 0.3923 | 44.95 | 0.3924 | 48.03 | 0.3366 |
| Any Deferred Earnings | 2022 | 45.43 | 0.0590 | 48.46 | 0.4798 | 46.54 | 0.5134 | 48.47 | 0.3989 |
| Any Foreign Born | 2018 | 20.12 | 0.0826 | 19.63 | 0.2344 | 20.12 | 0.2344 | 20.77 | 0.082 |
| Any Foreign Born | 2019 | 20.18 | 0.0374 | 18.46 | 0.2855 | 20.18 | 0.3125 | 21.49 | 0.039 |
| Any Foreign Born | 2020 | 20.48 | 0.0386 | 19.55 | 0.2798 | 20.47 | 0.3107 | 22.28 | 0.0399 |
| Any Foreign Born | 2021 | 20.57 | 0.0906 | 19.44 | 0.2855 | 20.57 | 0.2989 | 22.10 | 0.0898 |
| Any Foreign Born | 2022 | 20.69 | 0.0382 | 20.77 | 0.2789 | 20.70 | 0.296 | 21.99 | 0.0402 |
| Any Hispanic | 2018 | 15.06 | 0.0746 | 14.47 | 0.2340 | 15.32 | 0.1836 | 15.60 | 0.182 |
| Any Hispanic | 2019 | 15.05 | 0.0307 | 13.36 | 0.2604 | 15.61 | 0.2151 | 16.09 | 0.2178 |
| Any Hispanic | 2020 | 15.06 | 0.0381 | 13.64 | 0.2631 | 15.71 | 0.2751 | 16.15 | 0.2516 |
| Any Hispanic | 2021 | 14.93 | 0.0714 | 13.37 | 0.2374 | 15.39 | 0.2248 | 16.01 | 0.2228 |
| Any Hispanic | 2022 | 14.99 | 0.0419 | 14.10 | 0.2825 | 15.36 | 0.291 | 15.72 | 0.2786 |
| Any Non-Citizen | 2018 | 11.45 | 0.0646 | 11.03 | 0.1866 | 11.58 | 0.186 | 11.93 | 0.1576 |
| Any Non-Citizen | 2019 | 11.85 | 0.0329 | 10.26 | 0.2163 | 11.99 | 0.2488 | 12.72 | 0.1774 |
| Any Non-Citizen | 2020 | 12.24 | 0.0346 | 11.00 | 0.2308 | 12.37 | 0.2884 | 13.34 | 0.178 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Any Non-Citizen | 2021 | 12.43 | 0.0739 | 10.87 | 0.2116 | 12.28 | 0.2458 | 13.11 | 0.1773 |
| Any Non-Citizen | 2022 | 12.59 | 0.0320 | 11.93 | 0.2388 | 12.42 | 0.2818 | 13.18 | 0.1944 |
| Any White Non-Hispanic | 2018 | 67.99 | 0.1125 | 70.28 | 0.2932 | 67.43 | 0.2415 | 67.53 | 0.2222 |
| Any White Non-Hispanic | 2019 | 67.47 | 0.0469 | 73.66 | 0.3654 | 67.71 | 0.3343 | 67.69 | 0.3092 |
| Any White Non-Hispanic | 2020 | 66.79 | 0.0443 | 72.68 | 0.3558 | 67.10 | 0.3284 | 67.13 | 0.3109 |
| Any White Non-Hispanic | 2021 | 65.90 | 0.1028 | 72.53 | 0.3540 | 66.97 | 0.3044 | 67.24 | 0.3007 |
| Any White Non-Hispanic | 2022 | 65.63 | 0.0507 | 70.73 | 0.3885 | 66.18 | 0.3327 | 66.34 | 0.3273 |
| Income Greater Than 75th Percentile, Households with Children | 2018 | 25.00 | 0.1294 | 26.39 | 0.4698 | 25.05 | 0.4448 | 25.54 | 0.2405 |
| Income Greater Than 75th Percentile, Households with Children | 2019 | 25.00 | 0.0682 | 27.24 | 0.5902 | 25.37 | 0.5927 | 25.36 | 0.2718 |
| Income Greater Than 75th Percentile, Households with Children | 2020 | 25.00 | 0.0728 | 29.31 | 0.5275 | 25.00 | 0.5636 | 27.42 | 0.2656 |
| Income Greater Than 75th Percentile, Households with Children | 2021 | 25.00 | 0.1481 | 29.14 | 0.5270 | 25.12 | 0.4989 | 27.54 | 0.2817 |
| Income Greater Than 75th Percentile, Households with Children | 2022 | 25.00 | 0.0705 | 29.90 | 0.7076 | 25.61 | 0.6883 | 27.45 | 0.332 |
| Income Less Than 25th Percentile, Households with Children | 2018 | 25.00 | 0.1259 | 23.74 | 0.4355 | 25.18 | 0.4454 | 24.83 | 0.2425 |
| Income Less Than 25th Percentile, Households with Children | 2019 | 25.00 | 0.0662 | 21.64 | 0.4808 | 25.43 | 0.6008 | 23.89 | 0.2654 |
| Income Less Than 25th Percentile, Households with Children | 2020 | 25.00 | 0.0645 | 20.38 | 0.4813 | 24.73 | 0.5803 | 22.78 | 0.2633 |

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|---|---|---|---|---|---|---|---|---|---|
| Income Less Than 25th Percentile, Households with Children | 2021 | 25.00 | 0.1499 | 20.67 | 0.4798 | 24.80 | 0.5324 | 22.34 | 0.2517 |
| Income Less Than 25th Percentile, Households with Children | 2022 | 25.00 | 0.0651 | 21.25 | 0.5427 | 25.12 | 0.6424 | 23.68 | 0.3027 |
| Income Between 25th and 50th Percentile, Households with Children | 2018 | 25.00 | 0.1294 | 24.88 | 0.4045 | 24.76 | 0.4055 | 25.16 | 0.3029 |
| Income Between 25th and 50th Percentile, Households with Children | 2019 | 25.00 | 0.0599 | 24.84 | 0.4681 | 24.48 | 0.5536 | 25.62 | 0.3476 |
| Income Between 25th and 50th Percentile, Households with Children | 2020 | 25.00 | 0.0749 | 23.90 | 0.5002 | 25.27 | 0.5374 | 24.53 | 0.4114 |
| Income Between 25th and 50th Percentile, Households with Children | 2021 | 25.00 | 0.1476 | 24.32 | 0.5067 | 25.24 | 0.5432 | 25.13 | 0.3535 |
| Income Between 25th and 50th Percentile, Households with Children | 2022 | 25.00 | 0.0751 | 24.17 | 0.5578 | 25.04 | 0.6762 | 25.39 | 0.3754 |
| Income Between 50th and 75th Percentile, Households with Children | 2018 | 25.00 | 0.1286 | 24.99 | 0.4608 | 25.00 | 0.4603 | 24.47 | 0.2814 |
| Income Between 50th and 75th Percentile, Households with Children | 2019 | 25.00 | 0.0683 | 26.27 | 0.5271 | 24.72 | 0.5862 | 25.13 | 0.3415 |
| Income Between 50th and 75th Percentile, Households with Children | 2020 | 25.00 | 0.0746 | 26.40 | 0.5170 | 25.00 | 0.5418 | 25.27 | 0.3489 |

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| Income Between 50th and 75th Percentile, Households with Children | 2021 | 25.00 | 0.1475 | 25.87 | 0.4961 | 24.84 | 0.5117 | 24.99 | 0.3225 |
| Income Between 50th and 75th Percentile, Households with Children | 2022 | 25.00 | 0.0677 | 24.68 | 0.6146 | 24.23 | 0.6512 | 23.49 | 0.399 |
| SSA OASDI Child Benefits | 2018 | 2.58 | 0.0315 | 2.46 | 0.1156 | 2.54 | 0.125 | 2.59 | 0.1146 |
| SSA OASDI Child Benefits | 2019 | 2.50 | 0.0148 | 2.42 | 0.1245 | 2.58 | 0.1572 | 2.70 | 0.1379 |
| SSA OASDI Child Benefits | 2020 | 2.56 | 0.0142 | 2.38 | 0.1030 | 2.59 | 0.1315 | 2.75 | 0.1119 |
| SSA OASDI Child Benefits | 2021 | 2.33 | 0.0295 | 2.27 | 0.1159 | 2.51 | 0.1516 | 2.76 | 0.1295 |
| SSA OASDI Child Benefits | 2022 | 2.31 | 0.0165 | 2.24 | 0.1145 | 2.41 | 0.1448 | 2.69 | 0.1343 |
| Medicaid Receipt, Households with Children | 2018 | 57.59 | 0.1618 | 57.64 | 0.4867 | 57.59 | 0.4737 | 59.10 | 0.1622 |
| Medicaid Receipt, Households with Children | 2019 | 57.67 | 0.0711 | 57.50 | 0.5848 | 57.67 | 0.5955 | 60.09 | 0.071 |
| Medicaid Receipt, Households with Children | 2020 | 57.44 | 0.0735 | 54.46 | 0.5321 | 56.93 | 0.5274 | 57.44 | 0.2843 |
| Medicaid Receipt, Households with Children | 2021 | 56.65 | 0.1734 | 54.00 | 0.5715 | 56.65 | 0.6089 | 56.80 | 0.1733 |
| Medicaid Receipt, Households with Children | 2022 | 57.78 | 0.0829 | 55.47 | 0.5979 | 57.78 | 0.5716 | 58.53 | 0.0821 |
| SNAP Receipt, Households with Children | 2018 | 43.25 | 0.2327 | 41.91 | 0.7298 | 43.44 | 0.7638 | 43.69 | 0.2813 |
| SNAP Receipt, Households with Children | 2019 | 40.18 | 0.1120 | 40.22 | 0.8666 | 40.38 | 0.9103 | 42.72 | 0.1977 |
| SNAP Receipt, Households with Children | 2020 | 39.23 | 0.1155 | 37.42 | 0.9513 | 39.39 | 0.959 | 40.02 | 0.2075 |
| SNAP Receipt, Households with Children | 2021 | 38.63 | 0.2781 | 38.43 | 1.132 | 38.77 | 1.087 | 40.35 | 0.2892 |
| SNAP Receipt, Households with Children | 2022 | 34.47 | 0.2546 | 34.73 | 1.629 | 34.51 | 1.742 | 37.58 | 0.2564 |

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|---|---|---|---|---|---|---|---|---|---|
| SSA OASDI Disabled Benefits | 2018 | 7.18 | 0.0527 | 7.77 | 0.1889 | 7.18 | 0.1857 | 7.85 | 0.0529 |
| SSA OASDI Disabled Benefits | 2019 | 7.11 | 0.0268 | 8.63 | 0.2134 | 7.11 | 0.2463 | 8.68 | 0.0263 |
| SSA OASDI Disabled Benefits | 2020 | 6.99 | 0.0265 | 8.21 | 0.2086 | 6.99 | 0.2223 | 8.52 | 0.0265 |
| SSA OASDI Disabled Benefits | 2021 | 6.66 | 0.0504 | 8.15 | 0.2064 | 6.66 | 0.2212 | 8.61 | 0.0504 |
| SSA OASDI Disabled Benefits | 2022 | 6.31 | 0.0254 | 7.53 | 0.1925 | 6.31 | 0.2519 | 8.15 | 0.0255 |
| Dividend Income Receipt | 2018 | 29.77 | 0.0917 | 31.99 | 0.3402 | 30.68 | 0.3289 | 30.68 | 0.3077 |
| Dividend Income Receipt | 2019 | 30.35 | 0.0461 | 34.56 | 0.3571 | 32.13 | 0.3455 | 31.61 | 0.3367 |
| Dividend Income Receipt | 2020 | 30.70 | 0.0415 | 36.32 | 0.3667 | 32.38 | 0.3564 | 33.08 | 0.3453 |
| Dividend Income Receipt | 2021 | 30.11 | 0.0976 | 35.20 | 0.3685 | 31.62 | 0.3581 | 32.38 | 0.3596 |
| Dividend Income Receipt | 2022 | 35.18 | 0.0417 | 39.62 | 0.3864 | 36.32 | 0.3877 | 36.70 | 0.3655 |
| Matched to 1040 Record | 2018 | 80.21 | 0.0829 | 80.77 | 0.2501 | 80.44 | 0.2488 | 80.72 | 0.123 |
| Matched to 1040 Record | 2019 | 79.39 | 0.0433 | 81.63 | 0.2560 | 79.40 | 0.2914 | 81.74 | 0.1468 |
| Matched to 1040 Record | 2020 | 81.54 | 0.0365 | 83.47 | 0.2583 | 81.64 | 0.2571 | 83.90 | 0.1152 |
| Matched to 1040 Record | 2021 | 80.76 | 0.0668 | 83.59 | 0.2479 | 80.63 | 0.2472 | 84.28 | 0.1402 |
| Matched to 1040 Record | 2022 | 80.36 | 0.0387 | 82.73 | 0.3251 | 80.34 | 0.355 | 82.97 | 0.1447 |
| Medicaid Receipt | 2018 | 36.25 | 0.0879 | 35.52 | 0.3198 | 36.25 | 0.2894 | 37.28 | 0.0876 |
| Medicaid Receipt | 2019 | 36.16 | 0.0444 | 33.56 | 0.3413 | 36.16 | 0.3602 | 37.44 | 0.0444 |
| Medicaid Receipt | 2020 | 36.02 | 0.0478 | 32.08 | 0.3012 | 36.02 | 0.3285 | 36.08 | 0.0481 |
| Medicaid Receipt | 2021 | 35.74 | 0.0984 | 32.08 | 0.3253 | 35.74 | 0.3604 | 36.30 | 0.0989 |
| Medicaid Receipt | 2022 | 36.43 | 0.0447 | 33.96 | 0.3471 | 36.43 | 0.3859 | 37.85 | 0.0438 |
| SNAP Receipt | 2018 | 26.66 | 0.1260 | 25.72 | 0.4283 | 26.77 | 0.4235 | 27.50 | 0.153 |
| SNAP Receipt | 2019 | 24.56 | 0.0633 | 23.42 | 0.5010 | 24.66 | 0.5505 | 27.03 | 0.1096 |
| SNAP Receipt | 2020 | 24.08 | 0.0647 | 22.28 | 0.5057 | 24.17 | 0.5972 | 25.58 | 0.1113 |
| SNAP Receipt | 2021 | 24.11 | 0.1619 | 22.49 | 0.5482 | 24.18 | 0.6228 | 26.19 | 0.1693 |
| SNAP Receipt | 2022 | 20.61 | 0.1330 | 19.87 | 0.9453 | 20.62 | 1.053 | 22.41 | 0.1333 |
| 1098 Receipt (Mortgage Lender) | 2018 | 45.65 | 0.0926 | 45.83 | 0.3328 | 45.32 | 0.3039 | 44.46 | 0.2685 |

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|---|---|---|---|---|---|---|---|---|---|
| 1098 Receipt (Mortgage Lender) | 2019 | 44.24 | 0.0471 | 45.63 | 0.3843 | 44.85 | 0.3913 | 43.21 | 0.3903 |
| 1098 Receipt (Mortgage Lender) | 2020 | 44.51 | 0.0444 | 46.69 | 0.3931 | 45.80 | 0.4229 | 44.92 | 0.3809 |
| 1098 Receipt (Mortgage Lender) | 2021 | 40.28 | 0.0978 | 42.24 | 0.3620 | 41.29 | 0.3455 | 40.74 | 0.3195 |
| 1098 Receipt (Mortgage Lender) | 2022 | 44.97 | 0.0532 | 45.37 | 0.4847 | 45.26 | 0.4268 | 43.77 | 0.3643 |
| 1099-G Receipt | 2018 | 45.15 | 0.0873 | 44.65 | 0.3583 | 45.66 | 0.3426 | 45.47 | 0.3047 |
| 1099-G Receipt | 2019 | 42.37 | 0.0418 | 41.91 | 0.3502 | 42.86 | 0.3455 | 42.70 | 0.3101 |
| 1099-G Receipt | 2020 | 42.56 | 0.0500 | 41.60 | 0.3824 | 42.74 | 0.3715 | 42.82 | 0.3545 |
| 1099-G Receipt | 2021 | 49.64 | 0.1056 | 47.92 | 0.3825 | 49.73 | 0.3617 | 50.07 | 0.3387 |
| 1099-G Receipt | 2022 | 46.17 | 0.0508 | 44.51 | 0.4156 | 45.38 | 0.3914 | 45.34 | 0.3808 |
| 1099-MISC Receipt | 2018 | 21.53 | 0.0710 | 21.34 | 0.2792 | 21.38 | 0.2745 | 21.21 | 0.251 |
| 1099-MISC Receipt | 2019 | 21.73 | 0.0407 | 21.68 | 0.3053 | 22.05 | 0.3393 | 21.84 | 0.2826 |
| 1099-MISC Receipt | 2020 | 18.08 | 0.0385 | 18.28 | 0.3123 | 17.99 | 0.3301 | 18.44 | 0.2805 |
| 1099-MISC Receipt | 2021 | 8.11 | 0.0522 | 8.95 | 0.2269 | 8.26 | 0.2329 | 8.56 | 0.2129 |
| 1099-MISC Receipt | 2022 | 10.89 | 0.0320 | 11.74 | 0.2709 | 11.05 | 0.2917 | 11.16 | 0.2864 |
| 1099-R Receipt (Retirement) | 2018 | 38.31 | 0.0792 | 42.62 | 0.3181 | 39.61 | 0.3059 | 40.33 | 0.2919 |
| 1099-R Receipt (Retirement) | 2019 | 38.86 | 0.0394 | 46.35 | 0.3714 | 39.80 | 0.3588 | 40.91 | 0.3243 |
| 1099-R Receipt (Retirement) | 2020 | 39.55 | 0.0578 | 47.05 | 0.3753 | 39.76 | 0.3907 | 41.29 | 0.3707 |
| 1099-R Receipt (Retirement) | 2021 | 36.34 | 0.0895 | 42.76 | 0.3634 | 36.49 | 0.3414 | 38.03 | 0.3231 |
| 1099-R Receipt (Retirement) | 2022 | 39.67 | 0.0464 | 46.36 | 0.4166 | 40.11 | 0.409 | 41.23 | 0.3725 |
| Matched to Main Administrative Data | 2018 | 86.18 | 0.0744 | 87.17 | 0.2074 | 86.18 | 0.2095 | 86.87 | 0.0744 |
| Matched to Main Administrative Data | 2019 | 85.38 | 0.0388 | 88.96 | 0.2018 | 85.38 | 0.2328 | 88.38 | 0.0381 |

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| Matched to Main Administrative Data | 2020 | 86.55 | 0.0362 | 89.56 | 0.2015 | 86.55 | 0.2098 | 89.12 | 0.0361 |
| Matched to Main Administrative Data | 2021 | 85.64 | 0.0642 | 89.83 | 0.2119 | 85.64 | 0.2285 | 89.70 | 0.0639 |
| Matched to Main Administrative Data | 2022 | 85.90 | 0.0369 | 89.50 | 0.2488 | 85.90 | 0.2971 | 89.02 | 0.0386 |
| Interest Income Receipt | 2018 | 48.27 | 0.0975 | 49.80 | 0.3419 | 48.27 | 0.322 | 48.23 | 0.3015 |
| Interest Income Receipt | 2019 | 50.15 | 0.0516 | 53.86 | 0.4520 | 50.72 | 0.461 | 50.28 | 0.4255 |
| Interest Income Receipt | 2020 | 55.81 | 0.0494 | 60.06 | 0.3495 | 55.84 | 0.3969 | 56.57 | 0.3714 |
| Interest Income Receipt | 2021 | 53.16 | 0.1017 | 57.30 | 0.3248 | 53.32 | 0.3639 | 54.16 | 0.3493 |
| Interest Income Receipt | 2022 | 55.84 | 0.0530 | 59.58 | 0.4102 | 56.30 | 0.4701 | 56.70 | 0.4363 |
| Married (IRS Filing Status) | 2018 | 45.19 | 0.1103 | 46.63 | 0.3444 | 45.19 | 0.3014 | 45.45 | 0.1121 |
| Married (IRS Filing Status) | 2019 | 44.55 | 0.0509 | 47.50 | 0.3557 | 44.55 | 0.3706 | 45.68 | 0.0513 |
| Married (IRS Filing Status) | 2020 | 44.38 | 0.0490 | 47.26 | 0.3859 | 44.38 | 0.383 | 45.56 | 0.0486 |
| Married (IRS Filing Status) | 2021 | 43.12 | 0.1061 | 45.93 | 0.3215 | 43.12 | 0.3187 | 44.40 | 0.1065 |
| Married (IRS Filing Status) | 2022 | 43.46 | 0.0400 | 45.99 | 0.4325 | 43.46 | 0.3672 | 44.09 | 0.0403 |
| Rental Income Receipt | 2018 | 10.42 | 0.0533 | 10.94 | 0.2164 | 10.38 | 0.2127 | 10.37 | 0.204 |
| Rental Income Receipt | 2019 | 10.13 | 0.0267 | 10.89 | 0.2369 | 10.10 | 0.2327 | 10.03 | 0.2179 |
| Rental Income Receipt | 2020 | 9.75 | 0.0296 | 11.37 | 0.2432 | 10.12 | 0.223 | 10.45 | 0.2103 |
| Rental Income Receipt | 2021 | 9.21 | 0.0494 | 10.67 | 0.2243 | 9.55 | 0.2253 | 9.94 | 0.2232 |
| Rental Income Receipt | 2022 | 9.17 | 0.0289 | 10.25 | 0.2514 | 9.19 | 0.2516 | 9.18 | 0.2494 |
| SSA OASDI Retirement | 2018 | 28.16 | 0.0810 | 31.98 | 0.3102 | 28.18 | 0.207 | 29.11 | 0.0871 |
| SSA OASDI Retirement | 2019 | 28.09 | 0.0366 | 36.82 | 0.3483 | 28.09 | 0.2847 | 29.56 | 0.049 |
| SSA OASDI Retirement | 2020 | 28.24 | 0.0426 | 37.06 | 0.3576 | 28.30 | 0.272 | 30.14 | 0.0538 |
| SSA OASDI Retirement | 2021 | 28.67 | 0.0838 | 38.03 | 0.3615 | 28.73 | 0.2641 | 31.07 | 0.0867 |
| SSA OASDI Retirement | 2022 | 28.68 | 0.0401 | 37.91 | 0.4472 | 28.72 | 0.3235 | 31.30 | 0.0493 |
| Schedule A (Itemization) | 2018 | 42.47 | 0.0779 | 42.80 | 0.3272 | 42.18 | 0.2992 | 42.17 | 0.1847 |
| Schedule A (Itemization) | 2019 | 16.93 | 0.0389 | 16.41 | 0.2954 | 17.04 | 0.3231 | 16.14 | 0.1692 |
| Schedule A (Itemization) | 2020 | 15.97 | 0.0377 | 16.98 | 0.2959 | 16.20 | 0.3047 | 16.57 | 0.1787 |
| Schedule A (Itemization) | 2021 | 14.25 | 0.0708 | 15.27 | 0.2614 | 14.40 | 0.2522 | 14.76 | 0.1386 |
| Schedule A (Itemization) | 2022 | 13.58 | 0.0322 | 15.01 | 0.3036 | 13.74 | 0.3148 | 14.12 | 0.1595 |
| Schedule C (Business) | 2018 | 24.64 | 0.0812 | 23.84 | 0.2943 | 24.74 | 0.2886 | 24.41 | 0.1158 |

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|---|---|---|---|---|---|---|---|---|---|
| Schedule C (Business) | 2019 | 25.03 | 0.0469 | 23.16 | 0.3350 | 24.98 | 0.386 | 24.47 | 0.0963 |
| Schedule C (Business) | 2020 | 25.17 | 0.0450 | 24.14 | 0.3088 | 25.14 | 0.3513 | 25.95 | 0.0993 |
| Schedule C (Business) | 2021 | 24.38 | 0.0772 | 22.69 | 0.3401 | 24.38 | 0.3595 | 24.63 | 0.1137 |
| Schedule C (Business) | 2022 | 25.40 | 0.0389 | 23.95 | 0.3396 | 25.57 | 0.3694 | 25.32 | 0.1049 |
| Schedule SE (Self-Employment) | 2018 | 19.04 | 0.0768 | 18.47 | 0.2458 | 19.09 | 0.2435 | 18.87 | 0.1531 |
| Schedule SE (Self-Employment) | 2019 | 19.10 | 0.0390 | 17.61 | 0.2969 | 18.89 | 0.325 | 18.63 | 0.2034 |
| Schedule SE (Self-Employment) | 2020 | 19.04 | 0.0383 | 18.37 | 0.3112 | 19.05 | 0.3496 | 19.72 | 0.2159 |
| Schedule SE (Self-Employment) | 2021 | 17.82 | 0.0667 | 16.65 | 0.2850 | 17.82 | 0.322 | 18.06 | 0.1771 |
| Schedule SE (Self-Employment) | 2022 | 19.12 | 0.0363 | 18.22 | 0.2823 | 19.39 | 0.3287 | 19.14 | 0.2063 |
| Sectors 44-55 (Retail Trade) | 2018 | 22.09 | 0.0870 | 22.40 | 0.3234 | 22.22 | 0.3296 | 22.76 | 0.3074 |
| Sectors 44-55 (Retail Trade) | 2019 | 21.89 | 0.0399 | 22.54 | 0.3324 | 21.82 | 0.3782 | 23.22 | 0.3291 |
| Sectors 44-55 (Retail Trade) | 2020 | 21.95 | 0.0470 | 21.86 | 0.3375 | 21.45 | 0.3459 | 22.40 | 0.3266 |
| Sectors 44-55 (Retail Trade) | 2021 | 21.50 | 0.0986 | 22.40 | 0.3260 | 21.46 | 0.3414 | 23.26 | 0.2863 |
| Sectors 44-55 (Retail Trade) | 2022 | 22.23 | 0.0479 | 22.89 | 0.4023 | 22.25 | 0.4188 | 23.72 | 0.3799 |
| Sector 72 (Accommodation and Food Service) | 2018 | 17.05 | 0.0824 | 17.19 | 0.2766 | 17.16 | 0.2763 | 17.65 | 0.256 |
| Sector 72 (Accommodation and Food Service) | 2019 | 18.23 | 0.0432 | 18.13 | 0.3142 | 18.25 | 0.3697 | 19.60 | 0.3252 |

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| Sector 72 (Accommodation and Food Service) | 2020 | 18.39 | 0.0467 | 17.50 | 0.3072 | 17.96 | 0.3379 | 18.95 | 0.3118 |
| Sector 72 (Accommodation and Food Service) | 2021 | 16.68 | 0.0739 | 16.42 | 0.3199 | 16.57 | 0.3329 | 18.00 | 0.303 |
| Sector 72 (Accommodation and Food Service) | 2022 | 16.99 | 0.0426 | 17.02 | 0.3425 | 16.87 | 0.3964 | 18.12 | 0.3524 |
| SSA OASDI Spouse Benefits | 2018 | 5.05 | 0.0405 | 5.73 | 0.1515 | 4.93 | 0.1279 | 5.15 | 0.1222 |
| SSA OASDI Spouse Benefits | 2019 | 5.02 | 0.0194 | 6.48 | 0.1852 | 4.63 | 0.1455 | 4.93 | 0.1261 |
| SSA OASDI Spouse Benefits | 2020 | 4.91 | 0.0197 | 6.71 | 0.1768 | 4.87 | 0.1424 | 5.28 | 0.1189 |
| SSA OASDI Spouse Benefits | 2021 | 4.75 | 0.0399 | 6.54 | 0.1873 | 4.73 | 0.1553 | 5.14 | 0.1401 |
| SSA OASDI Spouse Benefits | 2022 | 4.56 | 0.0186 | 5.76 | 0.1840 | 4.37 | 0.1641 | 4.62 | 0.1478 |
| SSI Receipt | 2018 | 5.89 | 0.0480 | 6.05 | 0.1465 | 5.83 | 0.1524 | 6.37 | 0.0562 |
| SSI Receipt | 2019 | 5.86 | 0.0254 | 6.22 | 0.1816 | 5.85 | 0.222 | 6.96 | 0.0327 |
| SSI Receipt | 2020 | 5.71 | 0.0206 | 5.90 | 0.1616 | 5.64 | 0.1856 | 6.58 | 0.0337 |
| SSI Receipt | 2021 | 5.45 | 0.0471 | 5.82 | 0.1433 | 5.45 | 0.1851 | 6.62 | 0.0499 |
| SSI Receipt | 2022 | 5.29 | 0.0240 | 5.61 | 0.1881 | 5.28 | 0.2273 | 6.22 | 0.0301 |
| Matched to W-2 Record | 2018 | 76.89 | 0.0841 | 74.56 | 0.3143 | 77.18 | 0.2567 | 76.32 | 0.1494 |
| Matched to W-2 Record | 2019 | 76.64 | 0.0393 | 71.67 | 0.3893 | 76.83 | 0.3471 | 76.24 | 0.1481 |
| Matched to W-2 Record | 2020 | 76.56 | 0.0429 | 71.34 | 0.3488 | 76.67 | 0.3024 | 75.88 | 0.1592 |
| Matched to W-2 Record | 2021 | 75.82 | 0.0808 | 70.85 | 0.3340 | 75.92 | 0.2728 | 75.45 | 0.153 |
| Matched to W-2 Record | 2022 | 75.90 | 0.0430 | 70.84 | 0.4193 | 76.13 | 0.3244 | 75.35 | 0.1652 |
| WIC Receipt | 2018 | 7.86 | 0.0898 | 7.48 | 0.3022 | 8.07 | 0.3281 | 8.18 | 0.1068 |
| WIC Receipt | 2019 | 7.14 | 0.0523 | 6.31 | 0.3058 | 7.29 | 0.3688 | 7.82 | 0.0767 |
| WIC Receipt | 2020 | 6.86 | 0.0490 | 6.07 | 0.2761 | 6.94 | 0.3731 | 7.88 | 0.0675 |
| WIC Receipt | 2021 | 5.81 | 0.1017 | 5.34 | 0.2797 | 5.81 | 0.433 | 7.05 | 0.1184 |

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| WIC Receipt | 2022 | 7.17 | 0.2050 | 6.63 | 0.8931 | 6.90 | 1.614 | 8.29 | 0.3663 |
| SSA OASDI Widow Benefits | 2018 | 7.12 | 0.0422 | 8.35 | 0.1899 | 7.55 | 0.1667 | 7.66 | 0.1572 |
| SSA OASDI Widow Benefits | 2019 | 7.10 | 0.0261 | 9.34 | 0.2132 | 7.44 | 0.1726 | 7.58 | 0.1601 |
| SSA OASDI Widow Benefits | 2020 | 7.06 | 0.0275 | 9.42 | 0.2239 | 7.46 | 0.184 | 7.63 | 0.1671 |
| SSA OASDI Widow Benefits | 2021 | 6.96 | 0.0463 | 9.21 | 0.1977 | 7.48 | 0.166 | 7.74 | 0.1517 |
| SSA OASDI Widow Benefits | 2022 | 6.81 | 0.0277 | 8.65 | 0.2293 | 6.95 | 0.1924 | 7.31 | 0.1792 |

Source: ACS, SIPP, and administrative data. Table presents means of administrative data variable for different samples. The first column of estimates is our benchmark, which is a representative set of respondents and nonrespondents to the ACS. The other columns present estimates of the same administrative variable, but with SIPP respondents using different sets of weights. Comparing the latter SIPP columns with the ACS columns provides a measure of nonresponse bias for a given variable.

Table A.2: Change in Key Estimates

| Key Estimate | SIPP Interview Year | Original | S.E. Original | Adrec in NI and Rake | S.E. Adrec in NI and Rake | Adrec in NI only | S.E. Adrec in NI only | Adrec in Rake only | S.E. Adrec in Rake only |
|---|---|---|---|---|---|---|---|---|---|
| Median annual household earnings | 2018 | $42,117 | $491 | $43,590 | $473 | $43,090 | $392 | $43,220 | $452 |
| Median annual household earnings | 2019 | $41,491 | $411 | $42,600 | $384 | $41,720 | $537 | $42,920 | $490 |
| Median annual household earnings | 2020 | $45,112 | $610 | $45,670 | $483 | $45,000 | $446 | $46,600 | $472 |
| Median annual household earnings | 2021 | $42,820 | $473 | $43,350 | $359 | $42,470 | $493 | $44,440 | $499 |
| Median annual household earnings | 2022 | $46,863 | $632 | $47,590 | $599 | $46,910 | $513 | $47,340 | $651 |
| Median annual household income | 2018 | $58,656 | $444 | $59,540 | $303 | $59,300 | $254 | $59,040 | $324 |
| Median annual | 2019 | $61,707 | $530 | $62,460 | $439 | $61,890 | $438 | $62,920 | $559 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| household income | | | | | | | | | |
| Median annual household income | 2020 | $65,758 | $544 | $65,180 | $412 | $65,020 | $442 | $66,740 | $483 |
| Median annual household income | 2021 | $65,482 | $566 | $64,910 | $412 | $64,670 | $352 | $66,590 | $431 |
| Median annual household income | 2022 | $69,022 | $667 | $68,480 | $567 | $68,450 | $581 | $68,580 | $653 |
| Poverty rate | 2018 | 13.74% | 0.23% | 13.48% | 0.21% | 13.53% | 0.21% | 13.68% | 0.22% |
| Poverty rate | 2019 | 12.40% | 0.33% | 12.36% | 0.28% | 12.59% | 0.29% | 12.17% | 0.30% |
| Poverty rate | 2020 | 11.48% | 0.29% | 11.68% | 0.25% | 11.73% | 0.24% | 11.16% | 0.25% |
| Poverty rate | 2021 | 12.03% | 0.27% | 12.29% | 0.25% | 12.40% | 0.24% | 11.78% | 0.25% |
| Poverty rate | 2022 | 12.52% | 0.31% | 12.95% | 0.30% | 12.68% | 0.27% | 12.86% | 0.30% |
| Percent of households receiving means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) | 2018 | 25.79% | 0.26% | 25.34% | 0.22% | 25.02% | 0.23% | 25.46% | 0.25% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of households receiving means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) | 2019 | 27.57% | 0.33% | 27.31% | 0.26% | 26.57% | 0.28% | 27.93% | 0.28% |
| Percent of households receiving means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) | 2020 | 26.22% | 0.30% | 26.85% | 0.26% | 25.94% | 0.25% | 27.00% | 0.25% |
| Percent of households receiving means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) | 2021 | 27.38% | 0.31% | 27.88% | 0.23% | 26.98% | 0.25% | 28.08% | 0.26% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of households receiving means-tested benefits (Medicaid, SNAP, SSI, GA, TANF, or WIC) | 2022 | 28.30% | 0.36% | 28.27% | 0.27% | 27.46% | 0.31% | 29.45% | 0.29% |
| Percent receiving means-tested transfer income (SNAP, SSI, TANF, WIC, and GA) | 2018 | 13.73% | 0.20% | 13.34% | 0.19% | 13.25% | 0.18% | 13.47% | 0.19% |
| Percent receiving means-tested transfer income (SNAP, SSI, TANF, WIC, and GA) | 2019 | 14.44% | 0.30% | 13.69% | 0.22% | 13.89% | 0.24% | 13.98% | 0.25% |
| Percent receiving means-tested transfer income | 2020 | 12.79% | 0.26% | 12.38% | 0.22% | 12.58% | 0.22% | 12.22% | 0.22% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| (SNAP, SSI, TANF, WIC, and GA) | | | | | | | | | |
| Percent receiving means-tested transfer income (SNAP, SSI, TANF, WIC, and GA) | 2021 | 13.86% | 0.27% | 13.45% | 0.22% | 13.60% | 0.21% | 13.21% | 0.23% |
| Percent receiving means-tested transfer income (SNAP, SSI, TANF, WIC, and GA) | 2022 | 14.24% | 0.32% | 13.74% | 0.28% | 13.75% | 0.27% | 14.01% | 0.28% |
| Percent receiving SNAP benefits | 2018 | 11.04% | 0.20% | 10.72% | 0.19% | 10.65% | 0.18% | 10.81% | 0.19% |
| Percent receiving SNAP benefits | 2019 | 11.45% | 0.29% | 10.87% | 0.22% | 11.06% | 0.25% | 11.05% | 0.25% |
| Percent receiving SNAP benefits | 2020 | 9.91% | 0.25% | 9.64% | 0.21% | 9.85% | 0.22% | 9.45% | 0.21% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent receiving SNAP benefits | 2021 | 11.48% | 0.26% | 11.11% | 0.21% | 11.32% | 0.21% | 10.85% | 0.22% |
| Percent receiving SNAP benefits | 2022 | 11.83% | 0.31% | 11.45% | 0.28% | 11.47% | 0.27% | 11.64% | 0.27% |
| Percent receiving SSI income | 2018 | 2.56% | 0.07% | 2.41% | 0.06% | 2.42% | 0.06% | 2.50% | 0.07% |
| Percent receiving SSI income | 2019 | 3.07% | 0.10% | 2.76% | 0.08% | 2.82% | 0.08% | 2.89% | 0.09% |
| Percent receiving SSI income | 2020 | 2.86% | 0.08% | 2.57% | 0.07% | 2.65% | 0.08% | 2.61% | 0.08% |
| Percent receiving SSI income | 2021 | 2.63% | 0.08% | 2.34% | 0.06% | 2.44% | 0.07% | 2.36% | 0.06% |
| Percent receiving SSI income | 2022 | 2.64% | 0.10% | 2.36% | 0.07% | 2.41% | 0.07% | 2.47% | 0.08% |
| Percent receiving income from social insurance programs (OASDI, VA benefits, WC, UC) | 2018 | 23.79% | 0.13% | 23.49% | 0.13% | 23.59% | 0.13% | 23.66% | 0.14% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent receiving income from social insurance programs (OASDI, VA benefits, WC, UC) | 2019 | 25.01% | 0.18% | 24.46% | 0.16% | 24.72% | 0.16% | 24.74% | 0.17% |
| Percent receiving income from social insurance programs (OASDI, VA benefits, WC, UC) | 2020 | 24.98% | 0.16% | 24.49% | 0.16% | 24.62% | 0.16% | 24.74% | 0.16% |
| Percent receiving income from social insurance programs (OASDI, VA benefits, WC, UC) | 2021 | 32.42% | 0.22% | 31.90% | 0.21% | 32.10% | 0.21% | 32.15% | 0.21% |
| Percent receiving income from social insurance programs (OASDI, VA | 2022 | 27.96% | 0.21% | 27.27% | 0.21% | 27.40% | 0.21% | 27.64% | 0.21% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| benefits, WC, UC) | | | | | | | | | |
| Percent receiving OASDI income | 2018 | 21.17% | 0.12% | 20.86% | 0.11% | 20.93% | 0.11% | 21.02% | 0.11% |
| Percent receiving OASDI income | 2019 | 22.44% | 0.15% | 21.90% | 0.14% | 22.10% | 0.14% | 22.19% | 0.15% |
| Percent receiving OASDI income | 2020 | 22.32% | 0.14% | 21.88% | 0.12% | 21.95% | 0.14% | 22.13% | 0.13% |
| Percent receiving OASDI income | 2021 | 22.92% | 0.13% | 22.32% | 0.12% | 22.53% | 0.12% | 22.65% | 0.13% |
| Percent receiving OASDI income | 2022 | 23.03% | 0.16% | 22.32% | 0.14% | 22.49% | 0.15% | 22.70% | 0.14% |
| Percent covered by public health insurance | 2018 | 36.19% | 0.26% | 35.66% | 0.22% | 35.56% | 0.22% | 35.81% | 0.23% |
| Percent covered by public | 2019 | 38.29% | 0.34% | 37.34% | 0.23% | 37.64% | 0.25% | 37.82% | 0.27% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| health insurance | | | | | | | | | |
| Percent covered by public health insurance | 2020 | 37.32% | 0.29% | 36.71% | 0.25% | 37.07% | 0.25% | 36.68% | 0.26% |
| Percent covered by public health insurance | 2021 | 38.65% | 0.31% | 38.02% | 0.27% | 38.39% | 0.27% | 37.80% | 0.29% |
| Percent covered by public health insurance | 2022 | 40.03% | 0.36% | 39.38% | 0.27% | 39.36% | 0.28% | 39.90% | 0.30% |
| Percent covered by Medicaid | 2018 | 20.62% | 0.26% | 20.21% | 0.21% | 20.02% | 0.21% | 20.32% | 0.23% |
| Percent covered by Medicaid | 2019 | 22.04% | 0.33% | 21.35% | 0.22% | 21.51% | 0.24% | 21.74% | 0.26% |
| Percent covered by Medicaid | 2020 | 20.74% | 0.28% | 20.44% | 0.26% | 20.64% | 0.25% | 20.31% | 0.26% |
| Percent covered by Medicaid | 2021 | 21.57% | 0.31% | 21.34% | 0.26% | 21.41% | 0.26% | 20.98% | 0.28% |
| Percent covered by Medicaid | 2022 | 22.84% | 0.35% | 22.45% | 0.27% | 22.30% | 0.28% | 22.97% | 0.29% |

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|---|---|---|---|---|---|---|---|---|---|
| Percent covered by Medicare | 2018 | 17.70% | 0.09% | 17.47% | 0.08% | 17.53% | 0.08% | 17.60% | 0.09% |
| Percent covered by Medicare | 2019 | 18.87% | 0.12% | 18.42% | 0.12% | 18.57% | 0.11% | 18.64% | 0.12% |
| Percent covered by Medicare | 2020 | 19.11% | 0.11% | 18.71% | 0.10% | 18.82% | 0.10% | 18.90% | 0.11% |
| Percent covered by Medicare | 2021 | 19.83% | 0.11% | 19.32% | 0.10% | 19.52% | 0.10% | 19.53% | 0.10% |
| Percent covered by Medicare | 2022 | 20.18% | 0.15% | 19.55% | 0.11% | 19.78% | 0.11% | 19.85% | 0.13% |
| Percent covered by private health insurance | 2018 | 65.65% | 0.31% | 66.15% | 0.26% | 66.31% | 0.27% | 65.91% | 0.28% |
| Percent covered by private health insurance | 2019 | 66.44% | 0.38% | 67.19% | 0.31% | 66.94% | 0.32% | 66.94% | 0.33% |
| Percent covered by private health insurance | 2020 | 66.59% | 0.34% | 66.89% | 0.32% | 66.79% | 0.30% | 67.11% | 0.31% |
| Percent covered by private | 2021 | 66.08% | 0.34% | 66.27% | 0.30% | 66.10% | 0.30% | 66.66% | 0.31% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| health insurance | | | | | | | | | |
| Percent covered by private health insurance | 2022 | 64.85% | 0.38% | 65.13% | 0.33% | 65.36% | 0.33% | 64.70% | 0.34% |
| Percent covered by employer-sponsored health insurance | 2018 | 54.71% | 0.31% | 55.25% | 0.31% | 55.35% | 0.31% | 55.02% | 0.33% |
| Percent covered by employer-sponsored health insurance | 2019 | 55.31% | 0.38% | 56.02% | 0.34% | 55.85% | 0.35% | 55.83% | 0.35% |
| Percent covered by employer-sponsored health insurance | 2020 | 56.30% | 0.34% | 56.48% | 0.36% | 56.47% | 0.34% | 56.70% | 0.35% |
| Percent covered by employer-sponsored health insurance | 2021 | 56.31% | 0.36% | 56.45% | 0.36% | 56.31% | 0.34% | 56.83% | 0.37% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent covered by employer-sponsored health insurance | 2022 | 54.49% | 0.41% | 54.69% | 0.37% | 55.02% | 0.35% | 54.27% | 0.38% |
| Percent uninsured | 2018 | 9.23% | 0.16% | 9.20% | 0.16% | 9.16% | 0.15% | 9.24% | 0.16% |
| Percent uninsured | 2019 | 8.09% | 0.19% | 8.12% | 0.18% | 8.08% | 0.18% | 7.99% | 0.18% |
| Percent uninsured | 2020 | 8.42% | 0.18% | 8.61% | 0.19% | 8.41% | 0.18% | 8.41% | 0.18% |
| Percent uninsured | 2021 | 7.97% | 0.19% | 8.21% | 0.20% | 8.13% | 0.17% | 8.05% | 0.18% |
| Percent uninsured | 2022 | 8.05% | 0.21% | 8.29% | 0.21% | 8.16% | 0.21% | 8.25% | 0.21% |
| Percent with a disability (age 1+) | 2018 | 19.51% | 0.18% | 19.19% | 0.17% | 19.20% | 0.18% | 19.40% | 0.19% |
| Percent with a disability (age 1+) | 2019 | 20.16% | 0.22% | 19.60% | 0.19% | 19.75% | 0.20% | 19.86% | 0.20% |
| Percent with a disability (age 1+) | 2020 | 20.10% | 0.21% | 19.66% | 0.17% | 19.91% | 0.18% | 19.72% | 0.18% |
| Percent with a disability (age 1+) | 2021 | 20.61% | 0.21% | 20.09% | 0.20% | 20.34% | 0.19% | 20.16% | 0.20% |
| Percent with a disability (age 1+) | 2022 | 21.83% | 0.27% | 21.33% | 0.22% | 21.46% | 0.21% | 21.59% | 0.24% |
| Median household net worth | 2018 | $103,942 | $1,734 | $109,900 | $1,779 | $108,100 | $1,694 | $106,800 | $1,597 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Median household net worth | 2019 | $97,948 | $2,182 | $109,800 | $2,244 | $102,300 | $2,124 | $111,500 | $2,470 |
| Median household net worth | 2020 | $118,024 | $2,073 | $123,800 | $2,459 | $114,600 | $2,374 | $129,500 | $2,457 |
| Median household net worth | 2021 | $141,125 | $2,999 | $143,100 | $2,574 | $134,900 | $2,732 | $152,400 | $2,784 |
| Median household net worth | 2022 | $167,273 | $3,516 | $174,200 | $2,886 | $167,700 | $2,815 | $172,700 | $3,032 |
| Percent with a retirement plan or account | 2018 | 42.49% | 0.24% | 42.80% | 0.21% | 42.97% | 0.21% | 42.62% | 0.23% |
| Percent with a retirement plan or account | 2019 | 42.03% | 0.28% | 42.39% | 0.27% | 42.56% | 0.26% | 42.30% | 0.30% |
| Percent with a retirement plan or account | 2020 | 43.77% | 0.28% | 43.52% | 0.23% | 43.95% | 0.24% | 43.60% | 0.26% |
| Percent with a retirement plan or account | 2021 | 47.70% | 0.27% | 47.33% | 0.27% | 47.92% | 0.26% | 47.28% | 0.28% |
| Percent with a retirement plan or account | 2022 | 48.82% | 0.33% | 48.88% | 0.30% | 49.37% | 0.29% | 48.27% | 0.31% |

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|---|---|---|---|---|---|---|---|---|---|
| Children ever born per 1,000 women age 15 to 50 | 2018 | 1,247 | 10 | 1,259 | 9 | 1,230 | 9 | 1,237 | 9 |
| Children ever born per 1,000 women age 15 to 50 | 2019 | 1,226 | 14 | 1,246 | 11 | 1,228 | 11 | 1,241 | 12 |
| Children ever born per 1,000 women age 15 to 50 | 2020 | 1,128 | 12 | 1,183 | 12 | 1,122 | 11 | 1,179 | 12 |
| Children ever born per 1,000 women age 15 to 50 | 2021 | 1,107 | 12 | 1,141 | 12 | 1,088 | 12 | 1,154 | 13 |
| Children ever born per 1,000 women age 15 to 50 | 2022 | 1,113 | 14 | 1,152 | 14 | 1,097 | 14 | 1,174 | 14 |
| Children ever born per 1,000 men age 15 to 50 | 2018 | 965 | 11 | 983 | 10 | 958 | 9 | 963 | 10 |
| Children ever born per 1,000 | 2019 | 945 | 13 | 974 | 12 | 951 | 12 | 968 | 12 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| men age 15 to 50 | | | | | | | | | |
| Children ever born per 1,000 men age 15 to 50 | 2020 | 901 | 12 | 942 | 13 | 897 | 12 | 943 | 12 |
| Children ever born per 1,000 men age 15 to 50 | 2021 | 873 | 12 | 899 | 12 | 864 | 12 | 914 | 12 |
| Children ever born per 1,000 men age 15 to 50 | 2022 | 870 | 15 | 912 | 14 | 867 | 13 | 931 | 14 |
| Percent of mothers with multi-partner fertility | 2018 | 18.83% | 0.28% | 18.66% | 0.26% | 18.62% | 0.26% | 18.79% | 0.27% |
| Percent of mothers with multi-partner fertility | 2019 | 18.93% | 0.36% | 18.66% | 0.35% | 18.84% | 0.35% | 18.82% | 0.35% |
| Percent of mothers with multi-partner fertility | 2020 | 18.42% | 0.37% | 18.54% | 0.35% | 18.37% | 0.34% | 18.22% | 0.33% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of mothers with multi-partner fertility | 2021 | 19.13% | 0.35% | 18.93% | 0.34% | 19.11% | 0.34% | 18.77% | 0.32% |
| Percent of mothers with multi-partner fertility | 2022 | 18.79% | 0.44% | 18.77% | 0.42% | 18.22% | 0.37% | 18.87% | 0.42% |
| Percent of fathers with multi-partner fertility | 2018 | 15.87% | 0.31% | 15.76% | 0.32% | 15.76% | 0.32% | 15.87% | 0.31% |
| Percent of fathers with multi-partner fertility | 2019 | 15.89% | 0.36% | 15.94% | 0.36% | 15.69% | 0.34% | 15.97% | 0.36% |
| Percent of fathers with multi-partner fertility | 2020 | 16.54% | 0.38% | 16.58% | 0.38% | 16.44% | 0.37% | 16.49% | 0.37% |
| Percent of fathers with multi-partner fertility | 2021 | 16.23% | 0.35% | 16.28% | 0.38% | 16.25% | 0.38% | 16.17% | 0.37% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of fathers with multi-partner fertility | 2022 | 14.48% | 0.37% | 14.59% | 0.41% | 14.37% | 0.39% | 14.69% | 0.41% |
| Percent of workers working two or more jobs | 2018 | 15.06% | 0.22% | 14.99% | 0.23% | 15.07% | 0.23% | 15.01% | 0.24% |
| Percent of workers working two or more jobs | 2019 | 17.67% | 0.31% | 17.72% | 0.29% | 17.91% | 0.28% | 17.63% | 0.30% |
| Percent of workers working two or more jobs | 2020 | 17.45% | 0.29% | 17.17% | 0.30% | 17.47% | 0.30% | 17.19% | 0.29% |
| Percent of workers working two or more jobs | 2021 | 16.75% | 0.28% | 16.66% | 0.24% | 16.90% | 0.25% | 16.58% | 0.24% |
| Percent of workers working two or more jobs | 2022 | 17.50% | 0.33% | 17.57% | 0.33% | 17.60% | 0.33% | 17.57% | 0.31% |
| Percent of workers who are self-employed | 2018 | 13.25% | 0.22% | 13.37% | 0.19% | 13.38% | 0.19% | 13.26% | 0.20% |
| Percent of workers who are | 2019 | 13.59% | 0.27% | 14.02% | 0.25% | 13.85% | 0.27% | 13.78% | 0.26% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| self-employed | | | | | | | | | |
| Percent of workers who are self-employed | 2020 | 13.99% | 0.28% | 14.14% | 0.23% | 13.95% | 0.23% | 14.21% | 0.24% |
| Percent of workers who are self-employed | 2021 | 13.33% | 0.26% | 13.74% | 0.26% | 13.45% | 0.25% | 13.68% | 0.27% |
| Percent of workers who are self-employed | 2022 | 13.87% | 0.29% | 14.20% | 0.28% | 14.03% | 0.27% | 14.18% | 0.30% |
| Percent of workers who worked full-time, year-round | 2018 | 62.16% | 0.30% | 62.40% | 0.29% | 62.27% | 0.30% | 62.39% | 0.29% |
| Percent of workers who worked full-time, year-round | 2019 | 61.80% | 0.35% | 61.92% | 0.33% | 61.93% | 0.33% | 61.84% | 0.32% |
| Percent of workers who worked full-time, year-round | 2020 | 61.26% | 0.33% | 61.28% | 0.32% | 61.19% | 0.31% | 61.19% | 0.31% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of workers who worked full-time, year-round | 2021 | 56.81% | 0.33% | 56.36% | 0.31% | 56.76% | 0.32% | 56.36% | 0.33% |
| Percent of workers who worked full-time, year-round | 2022 | 59.87% | 0.40% | 59.70% | 0.40% | 59.97% | 0.39% | 59.40% | 0.41% |
| Percent of workers who work from home at least one day per week | 2018 | 14.68% | 0.21% | 14.78% | 0.21% | 14.85% | 0.21% | 14.74% | 0.21% |
| Percent of workers who work from home at least one day per week | 2019 | 15.53% | 0.30% | 15.79% | 0.27% | 15.71% | 0.26% | 15.66% | 0.27% |
| Percent of workers who work from home at least one day per week | 2020 | 18.28% | 0.28% | 18.15% | 0.30% | 18.23% | 0.27% | 18.26% | 0.29% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of workers who work from home at least one day per week | 2021 | 29.73% | 0.34% | 29.45% | 0.33% | 29.86% | 0.31% | 29.56% | 0.33% |
| Percent of workers who work from home at least one day per week | 2022 | 28.75% | 0.37% | 28.65% | 0.37% | 28.94% | 0.37% | 28.50% | 0.38% |
| Percent of households who experienced a change in composition | 2018 | 15.45% | 0.23% | 15.38% | 0.22% | 15.51% | 0.22% | 15.60% | 0.23% |
| Percent of households who experienced a change in composition | 2019 | 14.94% | 0.34% | 14.73% | 0.31% | 14.75% | 0.32% | 14.84% | 0.30% |
| Percent of households who experienced a change in composition | 2020 | 15.80% | 0.31% | 15.65% | 0.28% | 15.82% | 0.27% | 15.80% | 0.27% |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Percent of households who experienced a change in composition | 2021 | 15.61% | 0.28% | 15.20% | 0.27% | 15.29% | 0.27% | 15.53% | 0.29% |
| Percent of households who experienced a change in composition | 2022 | 14.20% | 0.33% | 14.41% | 0.34% | 14.08% | 0.31% | 14.67% | 0.34% |

Source: 2018-2022 SIPP and administrative data. Table presents results on how the estimated value of SIPP key estimates change with adding administrative data to the model. The original estimates are shown in the first column. Three different sets of estimates with administrative data-based weights are shown in the other columns. The sets correspond to the results of adding administrative data to both the household noninterview adjustment and the second stage calibration (raking step), adding administrative data just to the household noninterview adjustment, and adding administrative data just to the second-stage calibration.